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Review

Ag-In-Ga-S Quantum Dots with Narrow Emission and Near-Unity Quantum Yield

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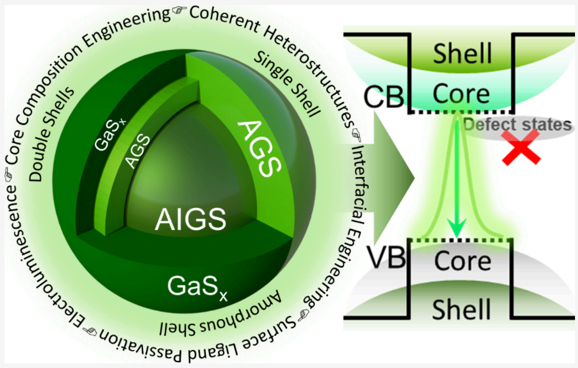
ABSTRACT: Ag-In-Ga-S (AIGS) quantum dots (QDs) have recently emerged as promising, environmentally benign emitters, offering a relatively high color purity and photoluminescence quantum yield (PLQY). Their quaternary composition provides precise control over the emission characteristics across the visible spectrum. Historically, the full potential of AIGS QDs was limited by pervasive bulk and surface defects, resulting in a broad and low-efficiency emission. Recent advances in core composition engineering, interfacial and heterostructure design, and surface-ligand engineering have propelled AIGS QDs into high-performance emitters with spectrally narrow emission and near-unity PLQY. This Review provides a critical examination of the evolution of AIGS QDs from their initial development to the latest breakthroughs, highlighting the key strategies for narrowing emissions, increasing brightness, defect suppression, and device implementation. Finally, the challenges and future opportunities for realizing high color purity and efficient electroluminescent devices are discussed for the proposed next-generation display and lighting technologies.



The next-generation displays have intensified the demand for visible light emitters with high brightness and exceptional color purity.[1] Among various luminescent materials, colloidal quantum dots (QDs) stand out as promising candidates owing to their narrow and vivid emissivity.[2] The advancement in chemical and colloidal synthesis routes has enabled QDs to exhibit facile color tunability across the visible spectrum. Their integration into QD-light-emitting diodes (QLEDs) has yielded remarkable external quantum efficiencies (EQEs) of 30.9, 28.7, and 21.9% for red, green, and blue spectral regions, respectively.[3−5] However, such QDs incorporate toxic heavy-metal elements, such as cadmium (Cd), which pose high-level concerns regarding environmental safety and regulatory compliance. This has prompted a critical research shift toward the development of Pb- and Cd-free QDs that align with stringent industrial standards and sustainability goals. III−V indium phosphide (InP) QDs emerged as a Cd and Pb-free candidate exhibiting tunable emission from green to red through synergistic effects of quantum confinement and the construction of core/shell heterostructures.[6,7] Nevertheless, achieving pure blue emission from the InP QDs remains a formidable challenge. Strategies such as extreme size reduction or alloying with wide-bandgap GaP have thus far only succeeded in producing azure-to-sky-blue region,[8] falling short of the deep blue required for wide color gamut. A further limitation lies in their intrinsically low molar absorption coefficient, which curtails their efficiency in color-by-blue conversion architectures by allowing significant leakage of the blue pump light, thereby compromising color purity[9] and making them unsuitable for color-by-blue-type devices. In addition to these issues, the synthesis of InP QDs is also very challenging due to the sensitivity of phosphorus precursors, which are highly dangerous and susceptible to degradation in ambient environmental conditions.[10]

I−III−VI_2 QDs have attracted enormous attention due to their unique compositional advantages.[11−14] The ternary (and quaternary) compositions provide an additional degree of freedom for emission tuning, effectively decoupling bandgap engineering from strict quantum confinement. This, along with their characteristically broad emission and high PLQYs, makes them promising for lighting applications and has been widely applied in white light-emitting diodes (WLEDs).[15] However, their broad emission spectrum, originating from radiative defects, has long hindered their adoption in displays that require narrow emitters. The pioneering studies by Kuwabata's

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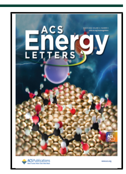

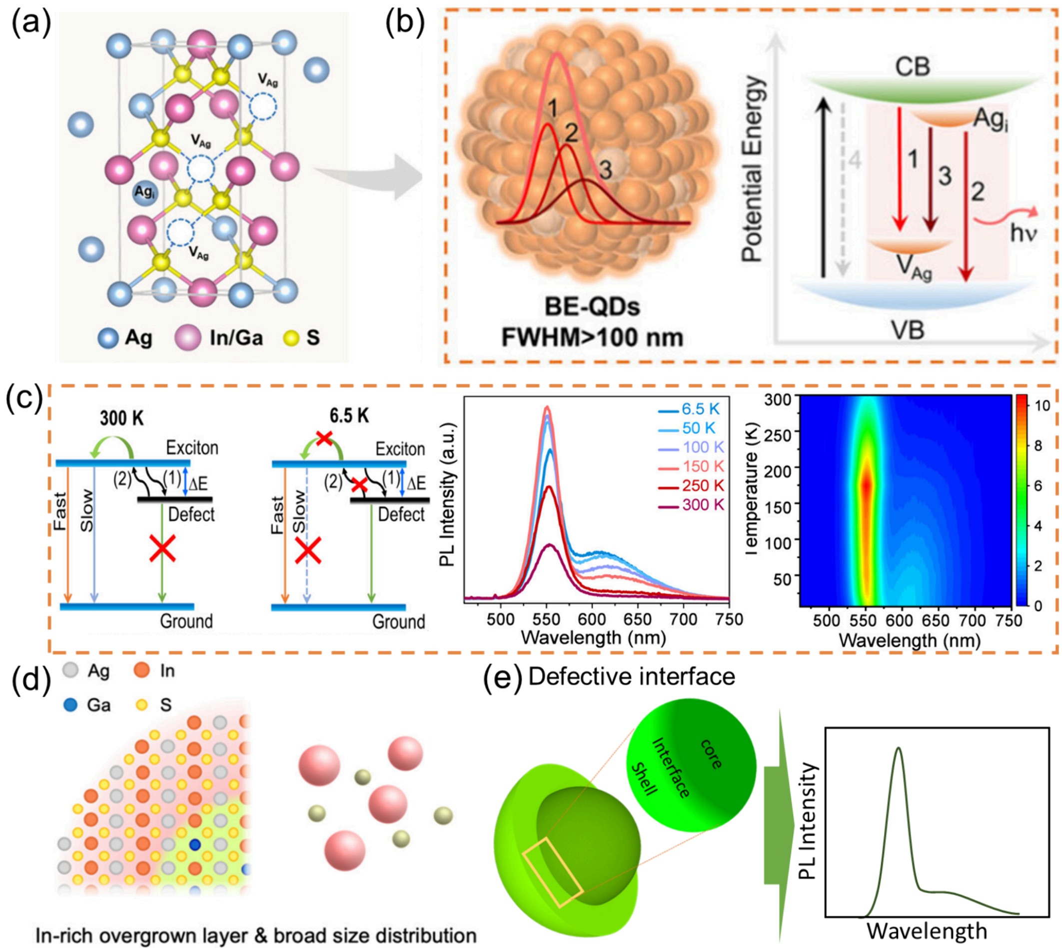

Figure 1. (a) Illustration of Ag-vacancy defects and the (b) relevant QD-scheme and defective band structure of the AIGS QDs. Reproduced with permission from ref 28. Copyright 2025 American Chemical Society. (c) Charge transition scheme, temperature-dependent PL spectra, and the corresponding pseudocolormap of AIGS QDs. Reproduced with permission from ref 26. Copyright 2025 American Chemical Society. (d) Scheme of the In-rich surface of AIGS QDs. (e) Scheme of AIGS-based core/shell structure exhibiting interface and the relevant PL spectrum demonstrating the defect emission tail. Reproduced with permission from ref 29. Copyright 2025 American Chemical Society.

group on I–III–VI_2 QDs enabled sharp band-edge emission, opening new avenues for engineering their optical properties and emission characteristics.[16,17] Recently, $AgInGaS_2$ (AIGS)-based QDs have garnered growing attention because these QDs achieve narrow-band emission via synergistic core compositional engineering and suitable surface passivation strategies, including Ga-based shell, coherent heterostructures, and surface ligand manipulations.[18,19] The effectiveness of these strategies lies in their collective ability to passivate surface defects and mitigate the unwanted radiative recombination pathways, providing a relatively narrow emission spectrum that is in high demand for next-generation wide-color-gamut displays.

Several recent reviews on heavy-metal-free QDs have primarily focused on their applications in solar energy conversion and general material development, emphasizing synthesis strategies and device architectures.[20,21] Some reviews are focused on I–III–VI-based QDs,[18] however, they lack the chemistry aspects of achieving narrow emission or their advances and limitations in the QLEDs. Here, we provide a comprehensive perspective on multinary AIGS QDs, highlighting compositional, interfacial, and shell engineering strategies that enable high PLQY, narrowed emission line widths, and improved color purity. By bridging materials design with device-relevant considerations, this work outlines a roadmap toward next-generation, environmentally friendly, high-performance emitters that are potentially compatible with Rec.2020 color standards based on emission bandwidth and chromaticity coordinates while recognizing that fully optimized device-level demonstrations remain an ongoing challenge. These insights have not been systematically addressed in previous reviews. Special emphasis is placed on the mechanistic insights and emerging synthetic strategies critical for the transition from the broad defect-dominated emission to narrow band-edge luminescence through coherent heterogeneous shell growth and interface engineering. The critical role of postsynthesis ligand exchange mechanisms in achieving near-unity PLQY is examined, with a detailed discussion of the underlying mechanism and its impact on optical performance. Furthermore, the photophysical merits of AIGS QDs are critically evaluated against state-of-the-art InP-based QDs, highlighting their strong potential as environmentally friendly candidates for next-generation displays. The discussion also includes the integration of AIGS-based QDs in QLEDs and their current challenges of low EQE outcomes. Finally, a concise summary with future perspectives and an outlook is provided to advance the development of AIGS QDs and their efficiency in QLEDs.

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■ ORIGINS AND NATURE OF DEFECTS IN AIGS QDS

Bulk Defects in Core

The optical properties of I−III−VI_2 QDs are generally defined by abundant defects, which are mainly radiative in nature and primarily responsible for their characteristically broad PL spectra.[18] These emissive states primarily originate from the bulk defects of the QDs including the localized donor−acceptor (DAP) recombination, a process fueled by the intrinsic lattice imperfections, including vacancies, interstitials, antisite defects, and structural irregularities inherent to their multinary composition.[22−25] Nag et al. investigated the origin of defect formation in AIGS QDs possessing a gradient composition, characterized by an In-rich inside core and Ga-rich toward the core surface.[26] First-principles calculations identified several Ag-related defects that exhibit low formation energies and play a critical role in governing the emission properties of these QDs, including V_{Ag}, Ag_i, Ag_{In}, Ag_{Ga}, In_{Ag}, and Ga_{Ag} (Figure 1a).[27]

Among these, shallow-level defects, V_{Ag} and Ag_i, are located near the valence band maximum (VBM) and conduction band minimum (CBM), typically functioning as DAP states. As depicted in Figure 1b (left), these states facilitate defect-mediated radiative recombination, leading to broad emission spectra. In a key complementary study, Zeng et al. demonstrated that the narrow emission spectra can be achieved through the elimination of such Ag-related structural defects.[28] They developed a synthesis route that promotes self-compensation for Ag-related defects during QD growth. Their analysis revealed that the unreacted Ag present in the reactor can fill V_{Ag} sites, while interstitial Ag_i atoms migrate at elevated temperatures to occupy remaining vacancy sites. This synergistic defect compensation effectively suppressed defect-mediated broad emission, thereby enhancing band-edge luminescence and yielding a high emission purity.

It is well-established that the broad PL spectrum with multiple split emission peaks after deconvolution corresponds to CBM-V_{Ag}, Ag_i-VBM, and Ag_i-V_{Ag} transitions, respectively (Figure 1b, right).[27,30] The recombination mechanism through these states is temperature-dependent. As shown in the PL and pseudocolormap spectra in Figure 1c,[26] the overall PL intensity decreases with increasing temperature, while a distinct defect-related emission band emerges below 200 K and intensifies upon further cooling. The spectral overlap between the excitonic emission tail and the defect-related emission indicates that these defects are shallow and located at a small energy offset (ΔE) below the lowest excitonic state. This minimal energy separation permits thermally activated back-transfer of charge carriers (transition (2) in Figure 1c) from the defect states to the excitonic states. At higher temperatures (200−300 K), this back-transfer is highly efficient, effectively suppressing defect-related emission within this range. In contrast, at temperatures below 200 K, the rate of back-transfer decreases substantially, allowing defect-mediated recombination to dominate the emission spectrum. The consequent reduction in exciton population leads to a quenching of the band-edge luminescence at a very low temperature (50 K), even though nonradiative decay is typically suppressed under such conditions. Below 50 K, the excitonic emission is further weakened as the enhanced suppression of process (2) severely limits the repopulation of excitonic states from the defect reservoir.

Collectively, this temperature-dependent relaxation behavior further confirmed the shallow nature of defect states in AIGS QDs and highlighted the critical role of thermal back-transfer in minimizing defect-mediated emission, which is a key prerequisite for achieving narrow-band emission at room temperature.

Surface and Interfacial Defects

Surface and interfacial defects play a decisive role in determining the optical properties of AIGS QDs due to their complex multicomponent chemistry. These defects typically arise from undercoordinated surface atoms, compositional inhomogeneity, and surface segregation of In, leading to the formation of shallow and deep trap states that promote radiative/nonradiative recombination and spectral broadening. In particular, interfacial defects at structurally imperfect core−shell boundaries are increasingly recognized as a key origin of tail emission in these QDs. A recent study identified that surface/interfacial defects promoting tail emission in AIGS QDs primarily originate from an In-rich surface, which induces a structurally imperfect core−shell interface, as illustrated in Figure 1d.[29] The synthetic pathway plays a decisive role in determining the resulting surface composition. One promising strategy employs an Ag−S−Ga$(oleate)_2$ molecular complex, which is subsequently alloyed with an In precursor.[31] This approach offers a competitive alternative to the conventional pathways where highly reactive Ag precursors first form Ag_2S intermediates, followed by cation exchange with In and Ga species.[32] Given that Ag_2S NPs exhibit off-stoichiometric compositions ($Ag_{2-x}S$),[33] the cation exchange process in such systems can introduce surface cation vacancies. These surface vacancies serve as DAP defect centers, giving rise to the observed low-energy tail emission. Therefore, precise control over the initial reaction stages critically influences the surface quality. For instance, employing an Ag−S−Ga$(oleate)_2$ molecular precursor route effectively suppresses the formation of off-stoichiometric $Ag_{2-x}S$ intermediates, thereby suppressing surface cation vacancies and associated DAP defects.[31] In contrast, conventional cation exchange pathways tend to generate more surface defects.[32] Additionally, high-temperature ripening has been shown to exacerbate In-rich surface and interfacial disorder (Figure 1e),[29] further deteriorating the luminescence properties.

A seed-mediated growth approach has been successfully employed to synthesize AIGS QDs, effectively addressing the surface defects associated with the phase segregation tendency between In^{3+} and Ga^{3+} ions.[34] Notably, this approach enables the preparation of high-quality QDs even under Ga-rich conditions, which conventionally promote surface trap states, defect emission and reduced PLQY. The resulting AIGS QDs exhibit enhanced emissive performance, achieving a narrow emission with fwhm < 40 nm. Through transient absorption spectroscopy (TA), the defect emission was identified as primarily arising from shallow surface states near the CBM, which were effectively suppressed using the seed-mediated synthesis approach. This method allows precise regulation of the Ag/In feeding ratio, facilitated by the strong coordination of 1-dodecanethiol (DDT) ligands toward soft acidic cations.

Synthetic Challenges and Intrinsic Limitations of AIGS QDs

The synthesis of multinary AIGS QDs presents inherent challenges due to the differing precursor reactivities among constituent elements. Such discrepancies often lead to

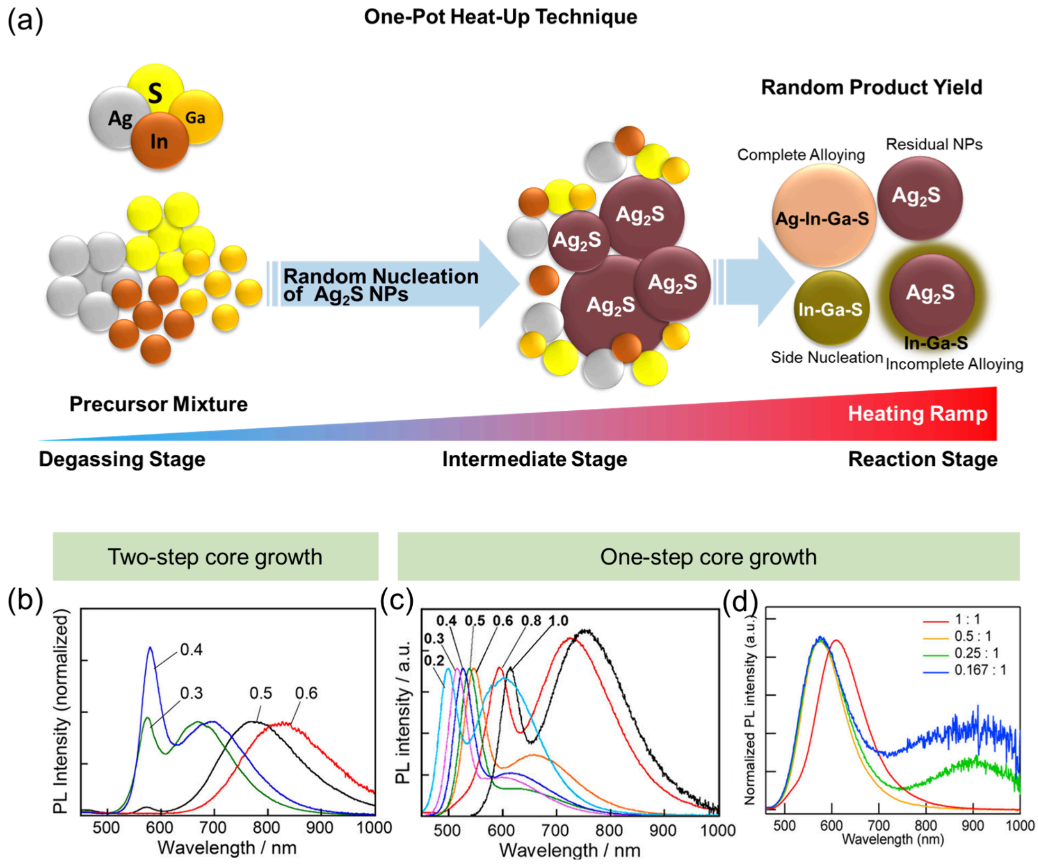

Figure 2. (a) Schematic illustration of intermediates formed during AIGS nucleation. Emergence of band edge emission in I–III–VI_2 QDs via (b) two-step heating and (c, d) one-step heating. Reproduced with permission from refs 17 and 46. Copyrights 2018 and 2021 American Chemical Society.

undesirable byproducts with inconsistent sizes and compositions, resulting in low-quality emissions. A persistent issue is the preferential binding of silver to sulfur relative to indium and gallium, which promotes the formation of silver sulfide (Ag_2S) intermediates during the initial stage.[35,36] Ligands such as alkanethiols can moderate their reactivity through the formation of stable metal-alkanethiolate complexes, which exhibit low reactivity compared to the unbound metal ions, thereby suppressing the formation of premature intermediates.[37−39] Group III precursors (typically halides, acetates, or stearates) tend to maintain or even enhance the reactivity of Group I elements. Consequently, the conversion of these metal precursors into sulfides typically requires a more nucleophilic sulfur source, such as a sulfur-amine mixture, bis(trimethylsilyl)sulfide, or ammonium sulfide, which offers stronger electron-donating capabilities than alkanethiols.[40,41] Although this strategy proves generally effective for synthesizing AIGS QDs, its inherent coupling of multiple reaction parameters inevitably restricts precise control over growth kinetics and product uniformity.

The synthesis of high-quality QDs is critically dependent on a well-defined nucleation process, which requires sufficient activation energy. Excessively suppressed precursor reactivity can therefore fundamentally complicate the synthesis mechanism. The intermediately formed Ag_2S NPs frequently act as essential intermediates or seed phases during AIGS QD nucleation,[36,39,42,43] significantly reducing the reactivity of the Ag precursor. It may slow the nucleation phase, leading to broader size distributions. According to the fundamentals of colloidal nanocrystal formation, achieving monodisperse QDs requires rapid nucleation within a narrow temporal window using highly reactive species.[44] Therefore, a conventional strategy that merely suppresses the most reactive component inherently conflicts with the fundamental requirements for controlled nucleation.

Although the band-edge emission of AIGS-based QDs largely arises from the introduction of GaS_x shells, the structural and size characteristics of the core QDs remain equally important. Ag-based QDs predominantly crystallize in either tetragonal or orthorhombic phases,[45] with the crystal structure proving decisive for optical performance. For instance, tetragonal-phase AIS QDs (a structural analog of AIGS) exhibited strong band-edge emission upon GaS_x shell coating,[16] whereas their orthorhombic counterparts exhibit markedly suppressed emission under identical shell-coating conditions. Although the underlying mechanism remained unclear, a reproducible synthetic route was developed for phase-selective tetragonal QDs. An optimized procedure using silver acetate (Ag(OAc)), indium acetate ($In(OAc)_3$) and thiourea at 140 °C successfully yielded tetragonal phased AIS QDs. However, attempts to moderate the reactivity of Ag through DDT complexation promoted the formation of Ag_2S byproducts, reducing the chemical yield. Extending to tetragonal-phased AIGS alloyed QDs by incorporating Ag(OAc), $In(OAc)_3$, and $Ga(DDTC)_3$ (gallium diethyldithiocarbamate) precursors in a coordinating solvent, the target tetragonal phase was achieved.[46,47] However, a significant amount of unwanted byproduct precipitates were formed, and

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the overall yield remained low. These results highlight the persistent challenges in managing precursor compatibility and reaction optimization during multinanosecond QD synthesis.

Mechanistic Insights and Emerging Synthetic Strategies

The synthesis of AIGS-based QDs is fundamentally governed by complex reaction kinetics and precursor chemistries rather than a direct nucleation pathway. A predominant mechanistic insight reveals that nucleation typically proceeds through the formation of chronic Ag_2S intermediates, owing to the higher reactivity of Ag-precursors with sulfur/thiol sources compared to their In/Ga-based counterparts (Figure 2a).[48] These Ag_2S intermediates initially serve as seed particles, facilitating cation exchange between Ag^+ and $(In/Ga)^{3+}$.[36,49] Subsequently, they can serve as catalytic media, accelerating the reaction of indium/gallium and sulfur precursors.[42] The catalytic pathway could lead to a rapid and uncontrolled increase in the average QD size, which is undesirable for obtaining monodisperse QDs with precisely regulated dimensions. A crucial advancement in controlling emission characteristics was demonstrated by Torimoto et al., who achieved band-edge emission from AIS QDs without prior shell growth by employing a two-step heating protocol during core growth.[17] This strategy involved an initial nucleation at a moderate temperature of 150 °C, followed by a high-temperature annealing step (>200 °C). The second heating step induced a substantial blueshift in the absorption spectra, indicating a compositional variation evolution and a reduction in the concentration of the narrow-bandgap Ag_2S intermediate. The PL spectra after the first heat treatment showed only a broad emission, arising from defect-related states/DAP recombination. This initial broad peak was progressively replaced by a sharper, blueshifted peak as the heating at higher temperatures progressed (Figure 2b). This two-step heating strategy thus enabled selective suppression of defect-related emission while promoting band-edge recombination, demonstrating a simple yet effective route to achieve high-purity emission from AIS cores without additional shelling.[17]

Extending this synthesis paradigm to quaternary AIGS QDs introduces greater complexity due to the additional elemental component.[50−52] The two-step synthesis used for AIS proved ineffective for AIGS QDs, as the initial low temperature (150 °C) was insufficient for the decomposition of gallium precursor.[17] Consequently, a single-step heating technique at elevated temperatures (e.g., 300 °C) is typically employed for the synthesis of quaternary AIGS QDs. Emission tuning is primarily achieved by modulating the Ag/(Ag + In + Ga) ratio from 0.3 to 0.6, with a nonstoichiometric value of 0.40, yielding dominant band-edge emission. These AIGS QDs crystallized into a chalcopyrite tetragonal structure, with crystalline phase intermediately located between those of the tetragonal AIS and AGS phases.[50−52] The absorption onset underwent a systematic blue-shift as the In/(In + Ga) ratio changed from 1.0 to 0.20, reflecting a widening of the bandgap with increasing Ga content, showing a composition-dependent behavior. A persistent challenge is the coexistence of narrow band-edge emission with broad defect-related emission, irrespective of the precursor ratio.[17,50−52] Both the band-edge and defect-related emission peaks showed an In/(In+Ga) ratio-dependent blueshift, consistent with bandgap widening through enhanced Ga incorporation.

A significant advancement in synthetic control was reported by Hoisang et al., who emphasized the critical role of precursor molecular design on the nucleation and growth processes of AIGS QDs.[46] They systematically compared two strategic approaches: (i) the use of separate precursors, e.g., metal-acetates with elemental sulfur or thiol/thiourea-based sources, and (ii) the use of single-source precursors utilizing metal-thiocarbamate complexes. In the first approach, $Ga(acac)_3$ was combined with *N*,*N*-dimethylthiourea (DMTU), which is a highly effective sulfur source for synthesizing size-controlled QDs with a strong PL intensity.[17] The second approach utilized $Ga(DDTC)_3$, a molecular complex formed by integrating both gallium and sulfur.[53] $Ga(DDTC)_3$ decomposes efficiently at moderate temperatures (∼120−150 °C) in the presence of amines,[53] which not only lowers the energy barrier for nucleation but also promotes more homogeneous incorporation of Ga^{3+} into the AIGS lattice. This controlled release of reactants mitigates defect formation by maintaining a favorable local chemical environment during crystal growth. Using $Ga(DDTC)_3$ in conjunction with Ag(OAc) and $In(OAc)_3$, AIGS QDs were synthesized with tunable In/Ga ratios at a relatively low temperature of 150 °C.[46] Subsequent GaS_x shell passivation yielded QDs exhibiting intense band-edge PL across multiple emission colors. Notably, a formulation with an In/Ga precursor ratio of 0.5:1 exhibited green emission with a PLQY of ∼59%. Comparative mechanistic studies using diethyldithiocarbamate complexes of indium and silver further highlighted the distinctive efficacy of $Ga(DDTC)_3$ in providing both gallium and sulfur species in a spatially and temporally coordinated manner, thereby enabling a low-temperature synthesis while maintaining a better optical quality. These results underscore the importance of precursor chemistry in tailoring the compositional homogeneity and emission characteristics of AIGS QDs for achieving narrow and pronounced band-edge emission.

Conversely, the conventional separate-precursor source strategy typically requires higher growth temperatures, leading to a broader size distribution, high defect densities, and consequently compromised PL performance. Indeed, QDs synthesized from a mixture of $Ga(acac)_3$ and DMTU at 150 °C exhibited signatures of monoclinic Ag_2S NPs, as depicted in Figure 2a, indicating that this temperature is insufficient for DMTU to decompose $Ga(acac)_3$ effectively and generate the intended metal sulfide. The stoichiometric variations played a substantial role in determining the tunable optical properties of AIGS QDs. For instance, systematic variation of the In/Ga precursor molar ratio induces a blueshift in the absorption onset as the indium content decreases, indicative of effective bandgap widening with increasing Ga incorporation.[46] However, this trend becomes less pronounced as the indium content is further reduced or the gallium content is increased. Under these conditions, an additional broad emission band emerges in the 700−1000 nm range. This new band appears alongside the band-edge peak and correlates with the absorption tails. These features are typically associated with residual Ag_2S NPs or incompletely converted AIGS nanoclusters (Figure 2c).[54] The PLQY exhibits a stoichiometric dependence, declining from ∼ 40% for the In/Ga ratio of 1:1 to ∼ 6% for 0.167:1 (Figure 2d). Although Ga-based surface passivation narrows the PL line width to 32−42 nm, the PLQY values remain modest, ranging between 28 and 59%, suggesting that higher gallium content in the core promotes defect states and induces nonradiative recombination pathways.[55,56]

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Uematsu et al. designed a synthesis route for homogeneously alloyed AIGS QDs without generating any intermediate precipitates.[32] This advancement surpassed the limitations of earlier methods that suffered from low yields due to undesirable intermediate byproducts. This strategy involves the rapid nucleation of Ag_2S intermediate NPs and their subsequent conversion into AIGS QDs via a single-step hot injection of Ag(OAc) stock solution into a premixed solution of dithiocarbamate complexes of In and Ga in a coordinating solvent. The resulting AIGS core QDs exhibited orange emission with a characteristic defect-mediated PL profile.[17,46,57] In contrast to the conventional one-pot heating of Ag(OAc), $In(OAc)_3$, and $Ga(DDTC)_3$ at 150 °C, which often leads to oversized Ag_2S-based byproducts,[46] the hot-injection strategy of Ag-source produced a clear colloidal solution with a broad defect-related emission profile.[32] The higher reaction temperature further enhances crystallinity while maintaining the tetragonal structure and is accompanied by a blueshift in both the absorption edge and defect-related PL bands. Subsequent Ga-based shell coating was carried out using $Ga(acac)_3$, DMTU, and $Ga(DDTC)_3$ as precursors.[46,47] To minimize self-nucleation of the shell material, the reaction temperature was rapidly heated to 230 °C and then gradually raised to 280 °C at a rate of 2 °C min^{-1}. The resulting AIGS-based core/shell QDs display a narrow PL peak, though significant defect emissions persist, particularly for cores synthesized at lower temperatures. Moreover, these QDs exhibit lower PLQYs than those prepared via the one-pot route using identical precursors,[46] suggesting subtle differences in compositional structure between the two synthetic approaches despite both yielding the tetragonal AIGS phase.

Mechanism of DDTC-Mediated AIGS Nucleation and Growth

Metal diethyldithiocarbamate (DDTC) complexes have emerged as effective precursors for the synthesis of AIGS QDs, primarily due to their ability to decompose at relatively low temperatures, thereby acting as nucleation regulators. Activated by amines, they efficiently generate metal sulfides through the release of thiocyanate derivatives.[53] In particular, the $Ga(DDTC)_3$ precursor undergoes Ga−S bond dissociation below 160 °C, releasing free ligands that subsequently coordinate with the injected Ag^+ ions to initiate nucleation. Upon introduction of the Ag^+ source, the reaction mixtures darken and exhibit negligible luminescence. The appearance of absorption tails extending beyond 700 nm suggests the possible formation of Ag_2S NPs. However, the crystallographic structure studies revealed no information related to the Ag_2S phase. Instead, the product is identified as an indium−gallium-sulfide composition containing a substantial amount of Ag. The process is characterized by instantaneous nucleation, which often leads to rapid precursor depletion and subsequent Ostwald ripening (where larger particles grow at the expense of smaller ones), resulting in polydispersity.[58,59] Unlike binary QDs, multinary I−III−VI_2 systems are especially prone to Ostwald ripening-induced compositional broadening, because their constituent cations exhibit markedly different solubilities, diffusion kinetics, and bonding strengths. During ripening, these disparities promote selective dissolution and redeposition of more mobile species (e.g., Ag^+), thereby amplifying compositional inhomogeneity in addition to size growth.[60] When the synthesis is conducted at low temperatures, Ostwald ripening is largely suppressed. Compositional studies revealed the deposition of indium−gallium-sulfide quasi-shells on the surface of Ag_2S NPs. These quasi-shells served as a diffusion barrier, effectively suppressing further particle growth, as schematically depicted in Figure 3.

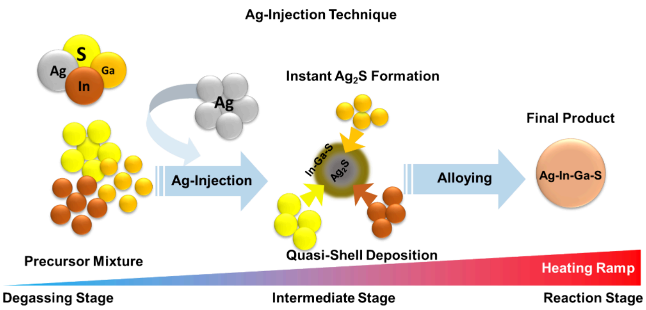

Figure 3. Scheme illustration of the synthesis of AIGS QDs.

Upon increasing the reaction temperature, the initial black dispersion gradually evolved into a reddish solution, accompanied by a distinct blue-shift in the absorption tail. Such a reddish crude mixture exhibited an orange PL, which was further enhanced upon annealing at higher temperatures due to the formation of homogeneous AIGS QDs. The In-to-Ga ratio, which governs the band gap, is also regulated during the formation of the Ag_2S/InGaS core/shell-like intermediate. A precision of the control and uniformity of the deposited composition is crucial for wavelength tuning. Substitution of $In(OAc)_3$ with $InCl_3$ was found to enhance the reaction kinetics, facilitated by more efficient ligand exchange among the precursor complexes.[32] As a softer Lewis acid, Indium halide ($InCl_3$) presented more favorable ligand exchange with $Ga(DDTC)_3$ than $In(OAc)_3$, producing partially substituted complexes such as $InCl_2$(DDTC), $InCl(DDTC)_2$, and their gallium analogues. As the total amount of In+Ga precursors exceeded that of Ag, an excess of indium−gallium sulfide formed relative to Ag_2S, subsequently transforming into phase-pure AIGS according to the reaction:

$$Ag_2S + In_xGa_{1-x}S \rightarrow Ag(In_xGa_{1-x})S_2$$

Even after complete conversion of Ag_2S into AIGS, residual partially alloyed indium−gallium sulfide remained in the reactor and passivated the surface of the underlying core as a quasi-shell. This core/shell-like configuration with In-Ga-S was mainly responsible for the narrow shoulder peak often observed in the PL spectrum of the AIGS core QDs. Although the ideal stoichiometry is very challenging, a key improvement in this work was the elimination of black-to-brown precipitates, which simplified postsynthetic purification and increased the product yield. In contrast to conventional approaches that suppress Ag reactivity, this method accelerated Ag_2S formation through the instantaneous injection of Ag^+ into a hot solution of $Ga(DDTC)_3$ maintained at 130 °C. At this temperature, $Ga(DDTC)_3$ partially decomposed to release [DDTC] anion, which acts as a highly efficient sulfur source.[32] These findings indicate that while metal-DDTC complexes regulate AIGS formation by supplying Ga and S precursors at low temperatures during nucleation, such conditions can increase the likelihood of premature Ag_2S formation, thereby deteriorating the composition of the resulting QDs.

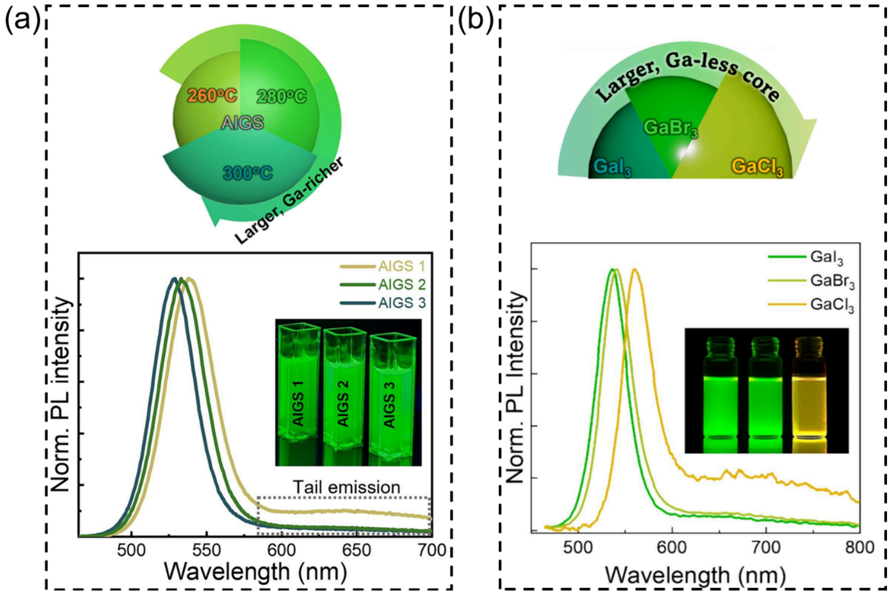

Figure 4. Regulating the synthesis of AIGS core QDs (a) by controlling the synthesis temperature. Reproduced with permission from ref 67. Copyright 2024 Elsevier. Regulating the synthesis of AIGS core QDs (b) by the Ga-precursor type. Reproduced with permission from ref 70. Copyright 2025 Elsevier.

Halide-Regulated Reaction Pathways

Halide ions (Cl^-, Br^-, and I^-) play a key role in the nucleation, growth, and surface chemistry of QDs across multiple material systems. Functioning as X-type ligands, halides preferentially bind to cation-rich facets, suppressing uncontrolled growth and promoting monodisperse nucleation.[61,62] In III−V systems such as InP, zinc halides ($ZnCl_2$, $ZnBr_2$) or indium halides ($InCl_3$) are added to reduce precursor reactivity through the stabilization of metal-halide complexes, enabling controlled formation of monomeric species and yielding narrow size distributions.[63,64] Halide additives also influence the multinary systems such as ZnSeTe by directing anion-exchange kinetics and preventing phase separation.[65] The strong coordination ability in a halides-mediated synthesis promotes "surface reconstruction" during early growth stages, reshaping lattice facets and modifying ion incorporation pathways.[66] These effects make halide-mediated chemistry a versatile strategy for achieving high-quality QDs with improved compositional homogeneity, narrower emission, and superior optoelectronic performance.

A halide-mediated synthesis leveraging iodide-based precursors was developed for AIGS QDs,[67] yielding QDs with high color purity and spectrally narrow emission. Considering the importance and sensitivity of the In/Ga ratio on the quality of core growth, the growth temperatures were adjusted to be between 260 °C (AIGS 1), 280 °C (AIGS 2), and 300 °C (AIGS 3) to tailor their optical properties (Figure 4a, top scheme). The average core diameter increased gradually with growth temperature. Regardless of the growth temperature, all cores exhibited a tetragonal crystal structure. Compositional analysis revealed that lower temperatures favored In-rich cores, whereas higher temperatures resulted in a Ga-rich composition. These reaction kinetics are different from syntheses employing metal−organic precursors. Based on elemental reactivity, the intermediate phase in this halide-mediated route was identified as AIS,[68] with higher core growth temperatures promoting the incorporation of Ga^{3+} into the AIGS lattice. As a result, the PL fwhm narrowed with increasing growth temperature, attributable to the suppression of vacancy defects led by Ga incorporation, thereby diminishing defect-related emission (Figure 4a, PL spectrum). The PL spectra further exhibited a slight blueshift with increasing core growth temperature, despite the increase in core size that would normally induce a red shift due to quantum confinement effects.[69]

The combination of various metal-halide or metal-acetate precursors plays a crucial role in achieving effective surface passivation during core growth (Figure 4b, top scheme).[70] Iodide ions, as key X-type ligands, play a vital role in suppressing the defect-related emissions through efficient surface passivation.[29,71]

Systematic studies on precursor ratios further revealed that an excess amount of iodine is essential for maximizing emissive efficiency and achieving high PLQY. Through controlled synthesis of AIGS QDs using various GaX_3 (X= Cl, Br, I) precursors, it was demonstrated that the nature of GaX_3 significantly affected both particle size and composition of QDs, owing to differences in halide steric hindrance and reactivity. Larger halide ions were found to promote the formation of Ga-rich cores with superior surface passivation capability, as observed from the absence of the defect emission in the longer wavelength region (Figure 4b, PL spectrum), facilitating more effective heteroepitaxial shell growth and enhanced emissivity. Upon shelling, GaI_3-based AIGS QDs exhibited a superior PLQY of 95% compared with 63% for $GaCl_3$-based counterparts.

Lowering the reactivity of I-B elements slowed down the nucleation and growth of AIGS, which adversely affected the uniformity and quality of the resulting QDs.[72,73] The overall reactivity of the system is a critical factor in enabling the rapid and uniform formation of high-quality QDs with narrow size distributions. Strategies that minimize byproduct formation while preserving a sufficient overall reactivity are therefore essential. The introduction of highly reactive III-A precursors can match the high activity of I-B and VI-A elements,

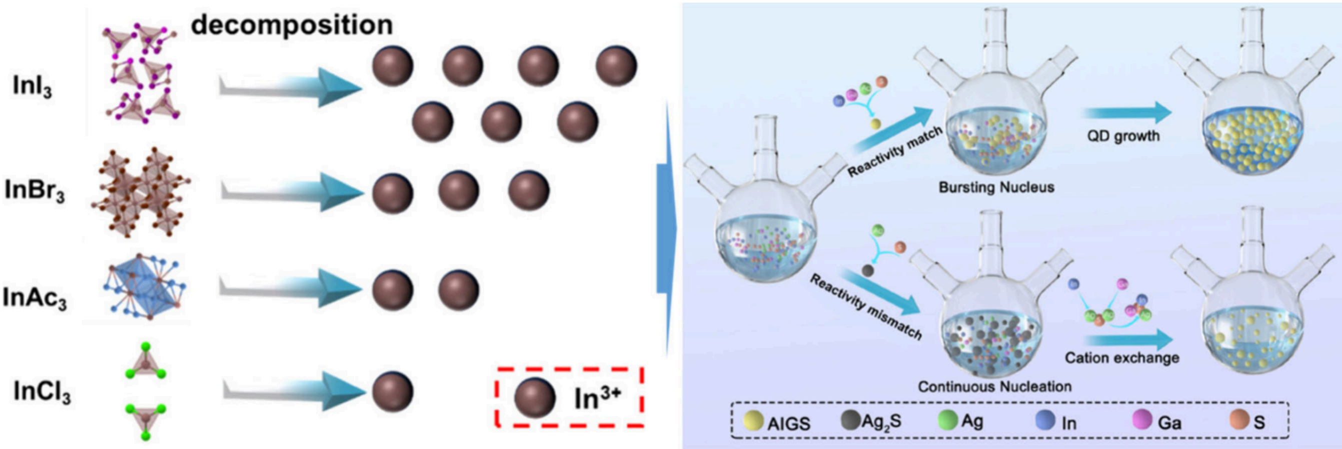

Figure 5. Illustration of indium precursor decomposition and supply of indium (left), reactivity matched vs unmatched synthesis of AIGS QDs. Reproduced with permission from ref 77. Copyright 2024 The Royal Society of Chemistry.

maintaining system reactivity and facilitating the synthesis of monodisperse AIGS QDs (Figure 5a,b).[74−76]

A multicomponent reactivity-matched synthesis approach produced uniform AIGS QDs with efficient luminescence features (PLQY reaching 90%).[77] Using highly reactive indium iodide (InI_3), with the capability of balancing the reaction speed of I-B, III-A, and VI-A elements while maintaining fast reactivity. This method promoted rapid nucleation, yielding monodisperse AIGS QDs with sufficiently high PLQY.[77] According to the HSAB principle, the differences in binding strengths between various anions (Cl, Ac, Br, and I) and metal cations can be exploited to select suitable indium precursors among $InCl_3$, $In(Ac)_3$, $InBr_3$, and InI_3. Indium cation, as a typical soft acid, exhibits the weakest interaction with soft bases, the iodide anion. The calculated formation energies of InI_3 were also relatively lower, implying higher instability and easier decomposition to release In^{3+} during the reaction (Figure 5). Consequently, InI_3 exhibits a much higher reactivity compared with the other indium sources for homogeneous core growth.

Hence, the emissive performance of AIGS QDs is intricately governed by the precursor chemistry, reaction kinetics, and surface composition. Surface defects, arising from incomplete conversion of intermediates, off-stoichiometric species, or In/Ga segregation, remain a primary source of broad defect-related emission. Strategic synthetic approaches, including seed-mediated growth, halide-mediated routes, and reactivity-matched precursor design, effectively mitigate these surface defects by promoting uniform element incorporation, suppressing cation vacancies in the bulk or at the surface, and enabling precise surface construction. These strategies collectively facilitate narrow band-edge emission, highlighting the critical interplay between surface engineering and synthetic control in achieving high-quality AIGS QDs.

HETEROSTRUCTURE DESIGN AND ROLE OF SHELL CHEMISTRY

Core/Shell Engineering Principles for AIGS

The prevalence of defect states in AIGS QDs is often linked to the inherent nonstoichiometry of the multinary composition, where the local atomic ratios within QDs deviate from their ideal chemical formula.[25] These defects can be mitigated by growing suitable shells, such as conventional Zn-based shells, on these QDs, significantly improving their PLQYs. However, the undesired interfacial alloying driven by Zn's high affinity with the core composition during shelling often leads to emission broadening.[78−81] Nevertheless, there is a decisive role of surface chemistry in dictating the balance between band-edge and defect-related emission. In particular, careful surface passivation and shell engineering have been shown to effectively suppress defect states, thereby enabling the emergence of narrow, well-defined band-edge PL. These insights highlight that while bulk or lattice defects contribute to the broad emission, the surface environment remains a critical factor in controlling the emission quality in AIGS QDs.

Amorphous GaS_x Shell: Surface Passivation and Limitations

In principle, a suitable shell material should minimize the possibility of interfering with core composition, such as alloying or forming solid solutions, while simultaneously ensuring efficient passivation and preventing the interfacial defect states.[82] However, the Zn-based shells grown on multinary QDs have exhibited typical broad emission in the visible region, suggesting their unsuitability for surface passivation.[16] For typical core/shell structures, the band alignment as well as the shell composition play a crucial role in efficiently confining the excitons.[83] For instance, in the case of AIS-based QDs with a bulk bandgap of ∼1.9 eV, indium sulfide (In_2S_3) composition with a slightly wider bulk bandgap of ∼2.2 eV satisfied a basic requirement for constructing a Type-I band alignment to effectively confine the carriers within the core. Investigation of the $Ag_2S-In_2S_3$ quasi-binary phase diagram revealed that the two systems are relatively immiscible, indicating rare chances for identical AIS system to form a homogeneous solid solution with In_2S_3.[84] Furthermore, because of the comparable elemental composition, indium is less expected to disrupt the AIS core composition compared to Zn-based shells,[85] making it a suitable alternative for achieving heterogeneous QDs with a well-defined core/shell interface. Consequently, the integration of In_2S_3 shell on AIS QDs led to a narrow PL peak emerging from band-edge recombination.[16] The intrinsic In_2S_3 NPs exhibited no luminescence, depicting their effective role as a passivation layer rather than the emitter. This optical response of core/shell AIS QDs underscored the intricate role of suitable shell composition/chemistry in tailoring the emission

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characteristics of multinary QDs. Nevertheless, the PL spectrum also included a significantly broad background emission, indicating the limited capability of the In_2S_3 shell for completely suppressing surface traps or DAP levels.[25]

A remarkable enhancement in emission monochromaticity was attained by substituting indium with gallium in the shell composition to form a GaS_x shell (Figure 6a).[86] The PL spectra after the GaS_x shell formation revealed a progressive suppression of the broad defect-related emission (>600 nm), which initially originated from the bare defective core. A narrow emission peak emerged on the shorter-wavelength side of the defect band, which was observed when the QDs were exposed to UV illumination (Figure 6b). Notably, the study based on the $Ag_2S-In_2S_3-Ga_2S_3$ quasi-ternary phase diagram supported a limited miscibility of the AIGS core with Ga−S shell elements.[87] Therefore, the substantial improvement in spectral purity was definitely a consequence of the larger bandgap of GaS_x (≈2.8 eV), which provided a stronger electronic confinement as compared to a marginally narrow band gap (≈2.2 eV) of the In_2S_3 shell. A pronounced reduction in the exciton relaxation lifetime components after the GaS_x shell growth was observed in comparison to the uncoated QDs, supporting the elimination of defect traps.[16]

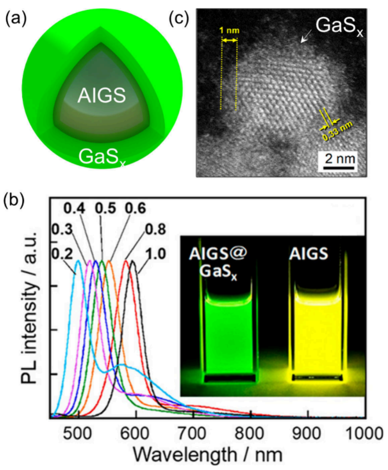

Figure 6. (a) Schematic diagram of the AIGS/GaS_x core/shell QD. (b) PL spectra of AIGS/GaS_x core/shell QDs and (c) HAADF-STEM image revealing the amorphous nature of the GaSx shell. Reproduced with permission from ref 17. Copyright 2018 American Chemical Society.

The Stokes shift provides a rough indication of the PL origin in QDs, a small Stokes shift typically suggests near-band-edge or excitonic emission, signifying a well-passivated surface and minimal trap-state involvement.[88,89] Whereas a large Stokes shift generally points to defective or deep trap-state-mediated emission, the carriers undergo substantial relaxation before radiative recombination. Thus, the magnitude of the Stokes shift can serve as a qualitative measure of the nature of emission.[88,89] The bare AIGS QDs synthesized from metal organic precursors normally exhibited a Stokes shift of 0.59 eV, which was significantly (about 10-fold) reduced after GaS_x shelling, consistent with the emission originating from the band-edge.[86] High-resolution transmission electron microscopy (HRTEM) images of core/shell QDs revealed an amorphous medium surrounding the crystalline QDs, indicating an amorphous nature of the GaS_x shell, as shown in Figure 6c.[46] The amorphous nature of GaS_x may be one of the reasons for the incomplete passivation of defects in AIGS QDs, resulting in a limited PLQY enhancement and a residual defect emission hump.[17]

The growth of GaS_x with the additives of $Ga(DDTC)_3$ and $GaCl_3$ as shell precursors instead of $Ga(acac)_3$ and DMTU/thiourea resulted in a highly crystalline shell region on AIGS QDs.[32] The addition of $GaCl_3$ during shell growth improved the optical characteristics of the resulting QDs, yielding a narrow PL with an fwhm of 31 nm. A preliminary addition of $GaCl_3$ to core QDs at ∼100 °C led to severe aggregation. This occurrence was prevented by injecting $GaCl_3$ at a higher temperature (160 °C), initiating ligand exchange between $Ga(DDTC)_3$ and $GaCl_3$ to form partial substitutes (e.g., $Ga(DDTC)_2Cl$), which bound to the core more effectively. This instantaneous protection preserved the core morphology and revealed no sign of amorphous structures surrounding the QD, leading to narrow emission. On the contrary, $Ga(acac)_3$ and DMTU-derived shell under the same procedure yielded a broader emission spectrum (fwhm ∼ 37 nm), highlighting that both the order of precursor introduction and the chemical nature of Ga and S sources influenced the QDs' surface morphology and the PL properties. The origin of the broad emission is primarily due to the presence of defects rather than the size disparity. It is noteworthy that, unlike the size-disparity induced broad emission in excitonic Cd- or InP-based QDs, the compositional disparity is mainly the cause of the broader emission in AIGS-based QDs.[43] The $GaCl_3$-engineered GaS_x shell improved the PL line width and PLQY, yet the defect emission in the longer wavelength region remained unavoidable, similar to the one depicted in Figure 6b. The optical properties of AIGS/GaS_x core/shell QDs are summarized in Table 1.

Lattice-Matched AGS Shell: Coherent Interface and Emission Narrowing

Amorphous GaS_x shells on AIGS QDs, while effective in reducing surface defects, exhibit several limitations in further enhancing the optoelectronic properties, thereby acting as a bottleneck for their development. Its disordered structure leads to ineffective carrier confinement within the core, resulting in broader emission line widths and reduced exciton lifetimes. Nonuniform GaS_x shell growth makes it hard to avoid strain or interfacial defects, strongly affecting the emission purity. Furthermore, the rigid chemistry of GaS_x restricts the tuning

Table 1. Summary of the Optical Properties of the AIGS/GaS_x QDs

No.	types of QDs	emission wavelength (nm)	PL fwhm (nm)	PLQY (%)	year	ref
1	AIGS/GaS_x	530	41	28	2018	16
2	AIGS/GaS_x	518	32−42	30−40	2021	46
3	AIGS/GaS_x-($GaCl_3$)	534	33−37	75−90	2023	32
4	AIGS/GaS_x	530	31	90	2024	77
5	AIGS/GaS_x	535	35	95	2025	70

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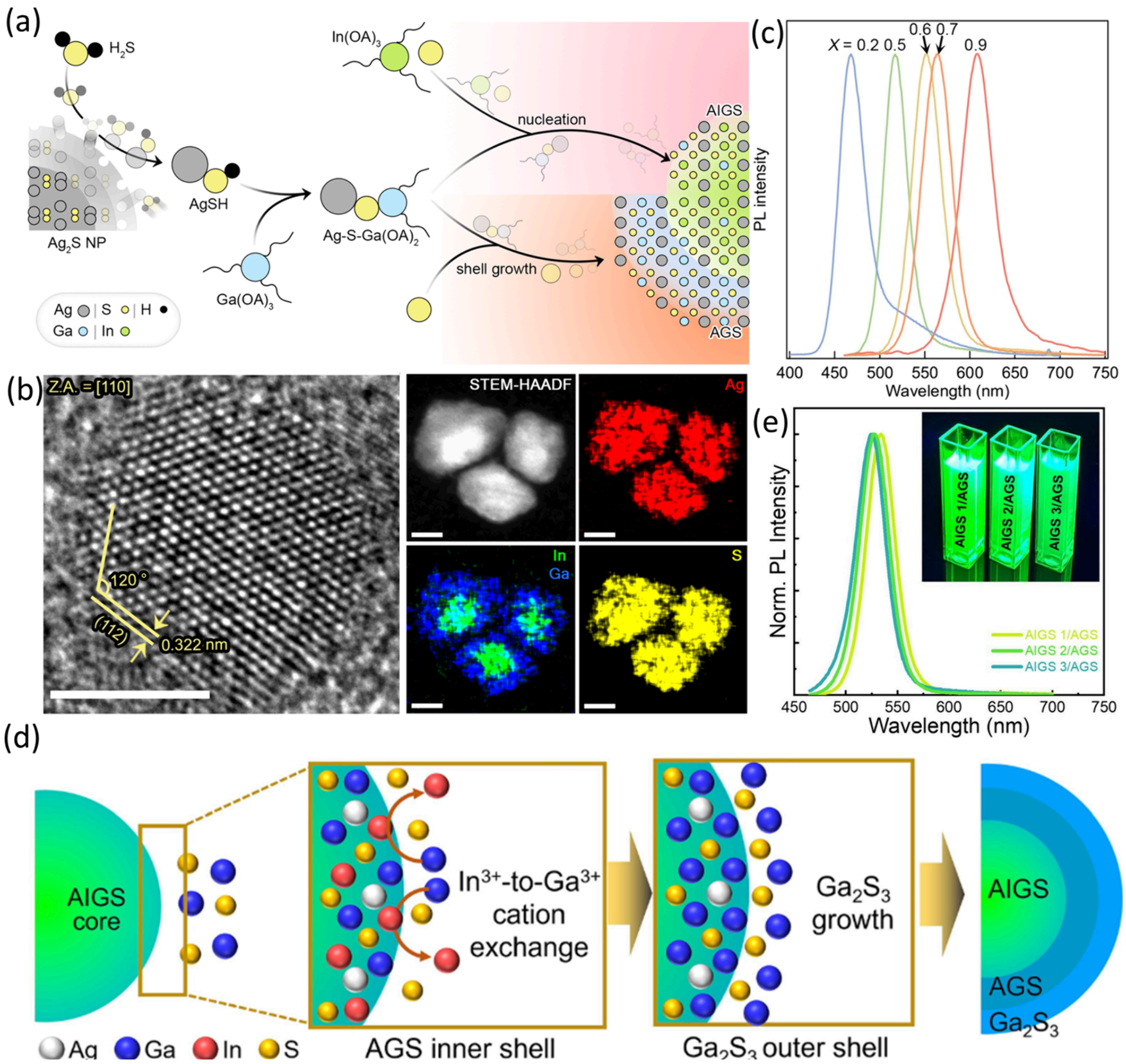

Figure 7. (a) Mechanism of Ag-S-Ga$(OA)_2$ formation and its subsequent conversion to AIGS and AGS shell. (b) HRTEM, STEM-HAADF and elemental mappings of AIGS/AGS QDs (scale bars: 5 nm). (c) PL spectra of the AIGS/AGS QDs. Reproduced from ref 31. Licensed under a CC Attribution 4.0, 2023 Nature. (d) Scheme illustrating cation-exchange induced AGS shell growth and (e) subsequent PL spectra of AIGS/AGS QDs. Reproduced with permission from ref 67. Copyright 2024 Elsevier.

of the interface for coherent core—shell structure, constraining precise control over optical and electronic properties as evidenced from the persistent tail emission despite near-unity PLQY.[32] In addition, the disordered nature of the GaS_x shell plays a detrimental role in the charge injection when integrated in the electroluminescence (EL) devices.[90] These factors collectively hinder the realization of AIGS QDs with optimal PL and EL performance.

To overcome the inherent limitations of the amorphous GaS_x shell, the development of a crystalline $AgGaS_2$ (AGS) shell has emerged as a promising strategy. The well-ordered lattice of AGS enables the formation of a coherent core—shell interface, thereby leading to a stronger carrier confinement and reduced interfacial strain, offering significant advantages over GaS_x-based shells. Additionally, AGS stands out as an ideal shell material for AIGS QDs due to its nearly identical crystal structure and symmetry, and a moderate lattice mismatch (~1.2%), while simultaneously forming a Type-I heterojunction that promotes efficient charge-carrier confinement within the core. However, to successfully grow an AGS-compositional shell, careful management of cation precursor reactivity is essential to prevent unwanted side reactions, such as nucleation of Ag_2S NPs or Ag precipitation. In this context, Bae et al. developed a synthetic route for the growth of AGS shell, wherein a tailored molecular precursor Ag-S-Ga$(OA)_2$ was prepared and utilized considering the difference in reactivity of various elements.[31] This stock complex not only served as a Ag, Ga, and S source but also resolved the individual reactivity imbalance issue. The Ag-S-Ga$(OA)_2$ stock solution was prepared by reacting Ag_2S NPs with *in situ*-generated H_2S, produced from a mixture of sulfur and alkylamine at temperatures above 130 °C, to yield soluble AgSH species. The AgSH species then reacted with Ga$(OA)_3$ to form the desired Ag-S-Ga$(OA)_2$ complex. The H_2S played a very crucial role in the formation of the Ag-S-Ga$(OA)_2$ complex. Injection of the In$(OA)_3$ precursor into the Ag-S-Ga$(OA)_2$ solution at 210 °C facilitated AIGS formation. The subsequent addition of the Ag-S-Ga$(OA)_2$ stock solution into the pregrown AIGS core dispersion resulted in AGS shell growth. A sulfur-rich reaction environment during the core/shell growth was found to raise the PLQY and suppress trap-related emission. The synthesis scheme is illustrated in Figure 7a.

Within this potential landscape, the flexibility in tuning the chemical composition of the AIGS core and the thickness of the AGS shell enabled the AIGS/AGS QDs to be bright, have remarkable photostability, and have broadly tunable emission (Figure 7c). The devised synthetic scheme allowed epitaxial AGS shell deposition, while maintaining the chalcopyrite crystal structure with a highly crystalline surface throughout the shell growth process, as displayed in Figure 7b. Notably, the broad defect-related emission characteristic of AIGS cores is markedly suppressed after AGS coating, as shown in Figure 7c. The thickness of the AGS shell played a crucial role in determining the PLQY, and a near-unity value was achievable at a shell thickness ranging from 1.6 to 2.9 nm with a PL fwhm of 30 nm. For AIGS/AGS QDs with higher gallium contents in the core, the PLQY decreased to around 50%, owing to inefficient carrier passivation by the AGS shell. The studies have also focused on the light emission capability of AIGS-based QDs at the single-particle level through single-dot spectroscopy.[17,31] The AIGS core with defects exhibited pronounced blinking behavior, whereas the AIGS/AGS QDs remained brightly emissive for a longer duration. This significant improvement indicated that in AIGS/AGS QDs, photogenerated carriers are well confined within the AIGS core under Type-I band alignment, favoring radiative recombination over surface trapping. A statistical analysis of multiple individual QDs showed that the average on-time fraction rose from 12% (for core-only) to 64% after coating with a 1.6 nm thick AGS shell. As the AGS shell became thicker, the half-lifetime of AIGS/AGS QDs correspondingly increased, indicating that the enhanced stability arose primarily from the expanded potential barrier width (i.e., shell thickness) rather than from a simple atomic-level surface passivation.

While the AGS shell procured a narrow emission line width with near-unity PLQY, a defect emission with a nominal magnitude remained persistent (Figure 7c). This emission likely arises from a combination of intrinsic bulk defects in the core and interfacial defects induced by strain at the core−shell interface, as commonly observed in various core/shell QD systems.[91] By adopting a cation-exchange strategy, Yang et al. achieved the controlled growth of an AGS shell on AIGS QDs, ensuring a smooth AIGS-to-AGS transformation that prevented interfacial stress between the core and shell, as illustrated in Figure 7d.[67] This method nearly resolved the tail emission issue in the emission spectra of the AIGS QDs (Figure 7e). The PLQY of resulting AIGS/AGS QDs in this case reached near-unity with the PL fwhm in the range 32−34 nm.

Interfacial Engineering Strategies

Interfacial engineering in core/shell QDs plays a pivotal role in optimizing their optical and electronic properties by controlling the lattice strain and suppressing defect formation at the interface. Carefully designed core−shell interfaces minimize nonradiative recombination pathways, leading to improved PLQYs and spectral purity.[92,93] Zheng et al. found that during the synthesis of AIGS cores using metal-dithiocarbamate complexes, the indium sulfide and its alloys, along with low-crystalline gallium sulfide, tend to deposit as an amorphous adsorbed layer (adlayer) on the surface of the AIGS cores.[94] This adlayer can form a weak interface for shell growth and lead to inferior luminescent performance of QDs due to existing surface and/or interface-related defects. The HF treatment has been widely reported to be effective for removing surface oxide-related defects in various QDs.[95,96] Inspired by such findings, Zheng et al. performed HF treatment to wash out the amorphous deposits on the surface of AIGS cores before shell growth that effectively removed the adlayer.[94] The HF-treated AIGS cores exhibited clear QD boundaries and well-resolved lattice fringes, confirming the effective removal of the surface adlayer through HF treatment and the fluorination of the surface via the formation of In−F and Ga−F bonds.[95,97] This treatment facilitated the formation of the AGS shell on core QDs with improved structural uniformity and better PLQY compared to the untreated QD counterpart. Following HF treatment of the core only, the PL peak of AIGS-HF QDs underwent a blue-shift, accompanied by a broadening of the fwhm to 155 nm and a decrease in PLQY. This reduction in emission efficiency was attributed to the formation of new surface defects resulting from the etched bare surface. The growth of the AGS shell transformed the broad emission into a narrow band-edge-derived emission (fwhm = 33 nm). The HF treatment marginally suppressed defect emission, resulting in green luminescence with a PLQY of 45%. These QDs exhibited enhanced chemical stability compared to the untreated counterpart and retained 75% of their initial PLQY after one month.

Lim et al. revealed that the instability of AIGS cores in coordinating solvents can induce crystallographic disorder at the core−shell interface during the shell growth, leading to the emergence of tail emission, commonly observed in AIGS-based core/shell QDs.[29] It was observed that AIGS cores are highly susceptible to Ostwald ripening in coordinating solvents, such as oleylamine (OLA), due to the strong coordination of OLA toward group-III elements.[98] This undesired process disrupts the chemical homogeneity of the ripened QDs because of the varied coordination strengths of the OLA with different metal cations, resulting in compositional inhomogeneity. Moreover, ripening promotes the formation of an orthorhombic phase, introducing both compositional and crystallographic disorders that generate DAP states at the AIGS/AGS interface.[99] Temperature-dependent PL analyses indicated thermal mixing between the band-edge electron state and a higher-lying donor state located approximately 20 meV above the CBM at 300 K. This disordered interface was effectively mitigated by introducing an iodide-based gallium source during synthesis, which suppressed Ga dissolution by weakening the coordination between the OLA and Ga cations (Figure 8a).[29] Consequently, the formation of a Ga-rich surface environment inhibited the formation of DAP states. The AIGS/AGS core/shell QDs synthesized with GaI_3 exhibited diminished tail emission as shown in Figure 8c. Although sufficient ligand coverage is essential for maintaining colloidal stability and regulating QD growth kinetics,[100−102] excessive coordination can promote dissolution or Ostwald ripening,[103−105] resulting in the formation of monomers or molecular species from pre-existing QDs.[106−108]

Temporal analyses of absorption and PL spectra revealed distinct activation thresholds for shell growth and ripening. As the temperature of OLA-dispersed AIGS cores increased without shell precursors, they displayed red-shifted absorption edges and significant tail emission, with the reduction in PLQY from ∼20% to ∼6% (Figure 8b). Meanwhile, core/shell QDs maintained their absorption edges and exhibited suppressed tail emission, a characteristic signature of AGS shell growth,

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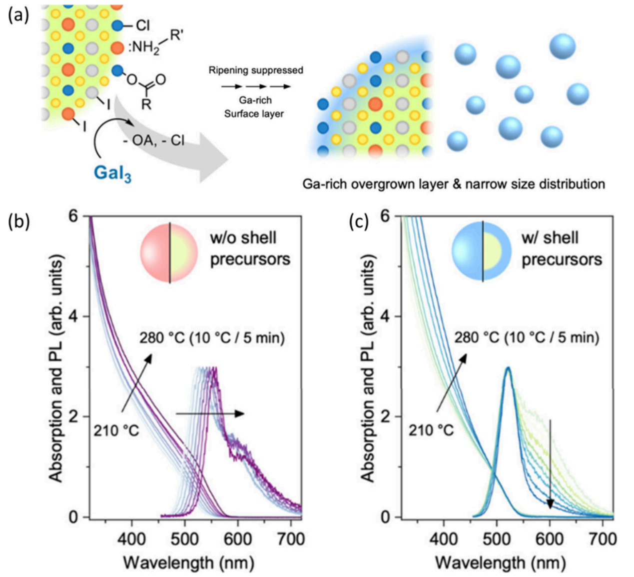

Figure 8. (a) Scheme defining the mechanism of halide-mediated surface treatment via GaI_3, preventing the Ostwald ripening, resulting in a homogeneous size distribution. (b) PL spectra of AIGS core QDs and (c) AIGS core QDs with shell precursors heated under similar conditions in OLA. Reproduced with permission from ref 29. Copyright 2025 American Chemical Society.

although a residual emission tail remained even with a PLQY of 75% (Figure 8c). The onset temperatures of ripening and shell growth indicated that ripening commenced approximately 40 °C earlier than the shell growth temperature, highlighting the thermal instability of AIGS cores in the OLA. In the case of ripened QDs, the elemental analyses by X-ray photoelectron spectroscopy (XPS) showed a reduction in the Ga 3d signal intensity and broadening in the In 4d peak, indicating an In-rich surface. This narrower band gap surface region promoted carrier delocalization and enhanced nonradiative exciton recombination. The incorporation of DAP states in this In-rich region further exacerbates nonradiative pathways. Different types of halide additives other than GaI_3, such as InI_3, $InCl_3$, and $GaCl_3$, with varying hard and soft acid−base combinations, were employed, among these, $InCl_3$ accelerated ripening and Ga removal, producing the giant QDs, the prominent redshift (200 meV), and the lowest Ga fraction (18%). InI_3 yielded similar results, whereas $GaCl_3$ caused moderate growth but retained more Ga, enduring greater stability, exhibiting the smallest redshift, and limited overgrowth in pristine cores. The optical properties of the AIGS/AGS core/shell QDs are summarized in Table 2.

■ POSTSYNTHETIC MODIFICATIONS AND SURFACE TREATMENTS

Alkyl Phosphine Ligand Exchange and Passivation Strategies

Postsynthesis surface treatments with alkyl phosphines, such as tri-*n*-octylphosphine (TOP), tributylphosphine (TBP), and tri-*n*-octylphosphine oxide (TOPO), have been highly effective in eliminating surface trap states and enhancing the PLQY of QDs. These soft Lewis base ligands exhibit a strong affinity for undercoordinated metal sites, particularly those of group-II and -III elements, thereby passivating surface dangling bonds and reducing nonradiative recombination pathways. Post-synthetic surface treatments with alkyl phosphines promote ligand exchange or reconstruction at the QD surface, leading to improved surface stoichiometry and suppression of defect-related emission. For instance, TBP treatment has been shown to restore the PLQY of degraded QDs from 8% to above 80% (10-fold enhancement) by repairing surface defects and enhancing charge-carrier confinement.[109] Similarly, TOPO and TOP have long been recognized as coordinating reagents that stabilize QDs' surfaces and minimize trap densities.[110,111] Overall, alkylphosphine-based surface treatments represent a simpler yet powerful strategy for trap removal, ensuring high PL efficiency and long-term optical stability of QDs. The post-treatment of AIGS-based QDs with the alkyl phosphine ligand family was found to enhance their band-edge PL intensity with suppressed defect component.[46,70] A TBP treatment of AIGS/GaS_x core−shell QDs showed a further boost in their PLQY. Figure 9a schematically illustrates that the QDs possess/develop surface-related traps that deteriorate their optical properties, and after the surface treatment with alkylphosphine ligands, their optical properties, were revived, and the surface traps were eliminated. Following TBP treatment, the band-edge emission intensity was significantly enhanced with a dramatic increase in PLQY from 38% to 68% (Figure 9b).[46] This enhancement led to a transition in emission color to pure green, owing to the quenched defect emission component (Figure 9b, right). Besides the surface passivation, TBP treatment was also effective in reviving the degraded AIGS/GaS_x QDs.[46] Without TBP, the band-edge emission of QDs stored in chloroform for 15 days weakened significantly, and the emission color shifted from green to yellow due to the emergence of strong defect-related PL, as shown in Figure 9c. The absorption spectra of the degraded QDs remained identical to those of the fresh samples, suggesting that the deterioration was mainly surface-related rather than due to structural/compositional changes. Upon addition of TBP, the band-edge PL intensity was substantially recovered, with a PLQY of 56%, exceeding that of the freshly synthesized QDs (38%) (Figure 9c, right).[46] The surface electron-accepting sites formed during long-term storage likely quenched the emission, and electron-donating ligands such as TBP effectively restored the optical properties.

Table 2. Summary of the Optical Properties of AIGS/AGS QDs

No.	type of QDs	emission wavelength (nm)	PL fwhm (nm)	PLQY (%)	year	ref
1	AIGS/AGS	518	55	77	2023	31
2	AIGS/AGS/GaS_x	526−534	33−34	95	2024	67
3	AIGS-HF/AGS	532	33	45	2024	94
4	AIGS-GaI_3/AGS		30	85	2025	29

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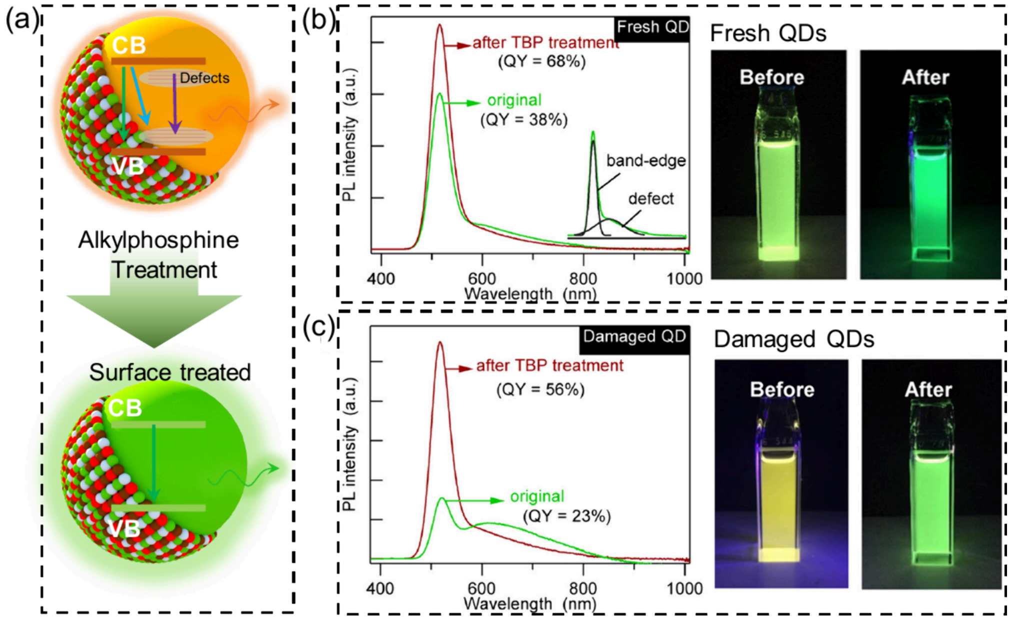

Figure 9. (a) Scheme illustrating alkyl phosphine surface treatment of QDs for improvement or restoration of PLQY. (b, c) PL spectra of AIGS QDs with and without TBP treatment for fresh and aged QDs. The right panel displays photographs of the QDs under UV illumination. Reproduced with permission from ref 46. Copyright 2021 American Chemical Society.

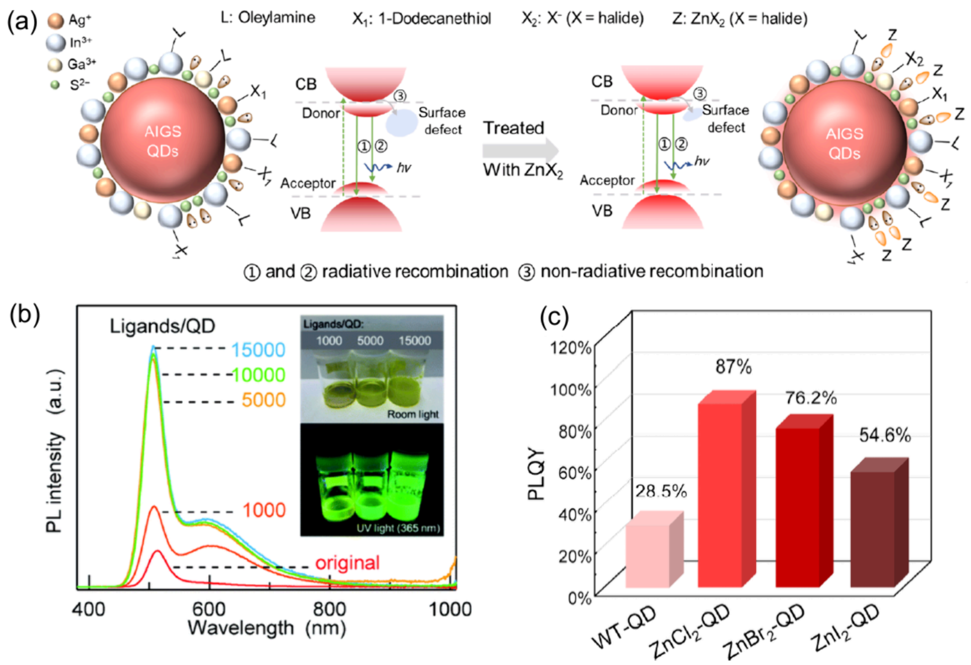

Figure 10. (a) Schematic illustration of Zn-type ligand passivated QDs surface, (c) comparison between the PLQYs of surface-treated QDs with ZnX_2 variants. Reproduced with permission from ref 119. Copyright 2024 Springer Publisher. (b) QDs dispersion showing PLQY enhancement with $ZnCl_2$ dosage. Reproduced with permission from ref 120. Copyright 2022 The Royal Society of Chemistry.

Surface Treatment with X- and Z-Type Ligands

The surface passivation using X- and Z-type ligands has emerged as an effective strategy to eliminate surface trap states and enhance the PLQY and the long-term stability of QDs. In the covalent bond classification (CBC) scheme, X-type ligands (such as halides, thiolates, and carboxylates) act as anionic species that bind to undercoordinated surface cations, while Z-type ligands (such as metal halides, metal carboxylates, or Lewis-acidic Zn complexes) coordinate to surface anions, thereby neutralizing surface charges and suppressing non-radiative recombination centers (Figure 10a). A balanced combination of X- and Z-type passivation effectively establishes the surface stoichiometry, reducing defect-induced trap states, and improves carrier confinement within the QD core. These strategies have been shown to significantly enhance PLQY, spectral stability, and environmental robustness across various II−VI and III−V QD systems.[112−114]

The AIGS-based QDs are prone to oxidation or ligand loss during purification, leading to trap formation and reduced PL efficiency. A halide-ion treatment of the core/shell QDs conducted by heating in the presence of a chloride ion source,

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Table 3. Comparison of the Optical Properties of AIGS, InP, and ZnSeTe QDs

No.	types of QDs	emission wavelength (nm)	PL fwhm (nm)	PLQY (%)	year	ref
1	AIGS/AGS	526−534	33	>90	2024	67
2	InP/ZnSeS/ZnS	508−535	35	>90	2022	125
3	ZnSeTe/ZnSe/ZnSeS/ZnS	509−530	44−47	80−90	2023	122

such as hydrochloric acid (HCl) or $GaCl_3$, substantially enhanced the PL performance.[32] The introduction of HCl as a concentrated aqueous solution into the crude QD dispersion, followed by dehydration under a vacuum at 100 °C, resulted in the formation of the OLA-hydrochloride. Then annealing at 260 °C significantly enhanced both band-edge color purity and PLQY, signifying the beneficial role of chloride ions in improving the optoelectronic properties of AIGS-based QDs.[115−117] In the context of surface chemistry, the chloride ion is generally classified as an X-type ligand; however, it can also function as a Z-type ligand. For instance, $GaCl_3$, when coordinated to a medium with an excess of Ga^{3+} species, acts as a Z-type ligand. Several studies have shown that Z-type ligands effectively enhance the PL intensity of chalcogenide and phosphide-based QDs.[114,118] Similarly, a notable PL enhancement in AIGS QDs following ZnX_2 treatment was also observed, with $ZnCl_2$ being highly efficient compared to other halide counterparts, as depicted in Figure 10c.[119] Although $GaCl_3$ treatment of AIGS/GaS_x core/shell QDs alone was ineffective under ambient conditions, the treatment performed at elevated temperatures resulted in significant increases in both band-edge PL intensity and PLQY (Figure 10b).[120] Hence, it was revealed that the chloride-ion treatment mechanism extends beyond simple surface passivation, likely involving partial diffusion of chloride ions into the GaS_x shell and the rearrangement of elements at the core−shell interface, thereby effectively passivating electronic trap states and enhancing the optical quality of the QDs.

■ OPTOELECTRONIC MERITS OF AIGS QDS

Comparison with InP and ZnSeTe QD Counterparts

Benchmarking AIGS QDs against their InP and ZnSeTe counterparts provides a comprehensive understanding of their relative structural, optical, and electronic characteristics. InP, as a representative III−V QD system, has achieved high PLQY (>90%) and narrow emission line widths (<40 nm) through meticulous control of core/shell interfaces, often utilizing lattice-matched ZnSe or ZnS shells to suppress nonradiative surface traps. However, the synthesis of InP QDs typically requires strict reaction control and highly reactive phosphorus precursors, which pose challenges for large-scale production and preserving surface stoichiometry.[121] On the other hand, ZnSeTe QDs, belonging to the II−VI family, benefit from relatively straightforward synthesis and broad compositional tunability through Se/Te ratio adjustment, yet they often suffer from structural inhomogeneity and phase segregation at high Te content, leading to broadened emission profiles and reduced spectral stability.[122]

AIGS-based QDs, feature exceptionally narrow emission line widths and near-unity PLQYs, surpassing InP[92,123−125] and ZnSeTe[126−128] QDs, both possessing comparable PL emissions, especially in the green spectral region. In particular, AIGS-based QDs exhibit an average single-dot emission line width of ~16.1 nm, which is narrower than that of InP (19.6 nm) and ZnSeTe (27.8 nm) QDs.[31] This superior spectral purity comes from the well-established synthesis route of AIGS QDs, leading to homogeneous composition with GaS_x and coherent AGS shells or both, forming a Type-I heterojunction, proving a strong carrier confinement. This confinement effect is further reflected in the ensemble PL spectrum (fwhm 30 nm), which is also considerably narrower than those of InP (36 nm)[92,123,125] and ZnSeTe (41 nm) QDs.[126]

Beyond narrow emission line widths, AIGS QDs demonstrate broad PL tunability across the visible spectrum from blue to red comparable to, and in some cases surpassing, InP and ZnSeTe systems.[31] Moreover, unlike InP and ZnSeTe systems, which rely heavily on Zn-based surface passivation, AIGS QDs exhibit superior interfacial stability with Ga-based shells due to their self-compensating composition and reduced susceptibility to interfacial strain. Collectively, these features position AIGS QDs as a promising class of environmentally friendly quantum emitters with the potential to surpass traditional InP and ZnSeTe QDs in terms of spectral purity, PLQY, tunability, and long-term shelf stability (Table 3).

High Molar Absorption Coefficient

AIGS QDs exhibit a remarkably high molar absorption coefficient (ε) of blue light, with a reported value of 1.24×10^6 M^{-1} cm^{-1}, which is further capable of increasing with QD volume.[122,129−132] These ε values are several-fold higher than those of InP and ZnSeTe QDs (Figure 11a), which are intrinsically limited in absorption, and comparable to the Cd-based QDs.[122] In color-by-blue device architectures, strong absorption of blue excitation light is critical for achieving high color conversion efficiency while minimizing blue light leakage. For instance, a QD plate developed by AIGS QDs, when illuminated by blue light, emits pure green color; the intensity of green emission increases with the amount of QDs, as shown in Figure 11b. The color-conversion devices based on AIGS and InP QDs are displayed in Figure 11c,d. Both types of QD plates were excited from the bottom by using blue light. As the QD concentration increased, blue light leakage decreased for both systems; however, this effect was significantly more pronounced in AIGS-based QDs, highlighting their superior molar absorption coefficient for blue light.[70,122] The exceptional absorption characteristics of AIGS QDs, combined with their high PLQY and tunable emission, underscore their strong potential as next-generation emitters and promising alternatives to conventional InP-based QDs.

Electroluminescent Application

Electroluminescent (EL) devices based on I−III−VI_2 QDs have rapidly advanced toward practical applications due to their heavy-metal-free composition and tunable emission across the visible range. Early implementations utilized broad-emitting I−III−VI_2 QDs, such as Ag-In-Zn-S and Cu-In-Ga-S, which enabled color-tunable output and the generation of white light for lighting applications.[133] For example, I−III−VI_2 broadband emitters have been integrated into white-light LEDs, leveraging blue excitation and down-conversion layers, which capitalize on their strong absorption and wide emission spectra.[134] More recently, attention has

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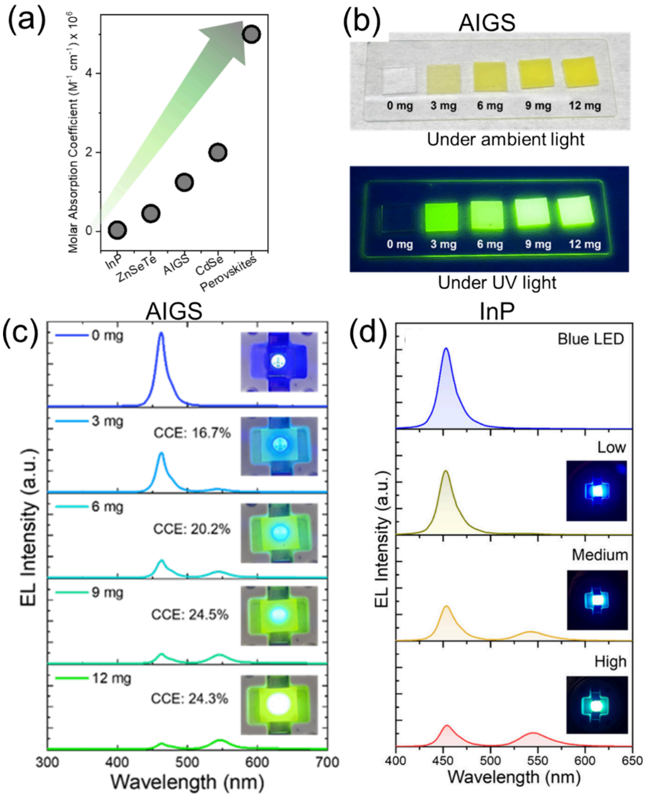

Figure 11. (a) Molar absorption coefficient of AIGS versus various QDs. (b) AIGS QDs slabs under ambient and UV light and (c) color-conversion devices based on AIGS QDs. Reproduced with permission from ref 70. Copyright 2025 Elsevier. (d) Color-conversion devices based on InP QDs. Reproduced with permission from ref 122. Copyright 2023 American Chemical Society.

shifted toward narrow-emitting color-tunable I–III–VI_2 QDs such as AIGS QDs for displays. Tsuzuki et al. fabricated QLEDs with AIGS QDs as the emitting layer (EML) that delivered a green EL spectrum.[90] Although the QDs exhibited vivid green PL with a sharp emission peak (fwhm of 32 nm), the corresponding QLEDs exhibited very poor EQEs and displayed EL spectra with pronounced defect-related emission, attributed to electron trapping at defect sites within the QD. These trap sites were mainly generated via the Ga-deficiency at the surface caused during the QD EML deposition. To resolve this issue, the QDs EML was treated with $GaCl_3$ and $Ga(DDTC)_3$ to eliminate such newly formed defects and provide efficient surface passivation to the EML. This treatment effectively compensated surface defect sites and enhanced the PLQY of deposited films from 16% to 32%. Additionally, $GaCl_3$ treatment of the ZnMgO electron transport layer (ETL) further improved the device performance by possible modulation of the relative band alignment of EML/ETL as well as suppressing direct electron injection into QD defect sites. As a result, the defect-related EL components were significantly quenched, leading to a narrow EL fwhm of 33 nm. Despite these advances, the device still exhibited poor charge carrier injection in the EML, characterized by the lower current density, suggesting either the inappropriate band alignment of AIGS/GaS_x QDs with ETL or the unfavorable nature of GaS_x shell for efficient charge injection.[135]

The shell plays a pivotal role in the efficient charge injection in the QDs, it can sometimes act as a barrier if the thickness is undesirably low or high, a balance is required.[136,137] Therefore, considering the limitation in GaS_x shell thickness control, the original QDs EML layer was modified via mixing with electron transport materials (ETMs) to improve the overall charge injection characteristics. By mixing TmPPyTz in AIGS/GaS_x QDs, the resulting QLED with modified EML demonstrated improved EL performance, featuring lower driving voltages and higher luminance, as demonstrated in Figure 12a. Furthermore,

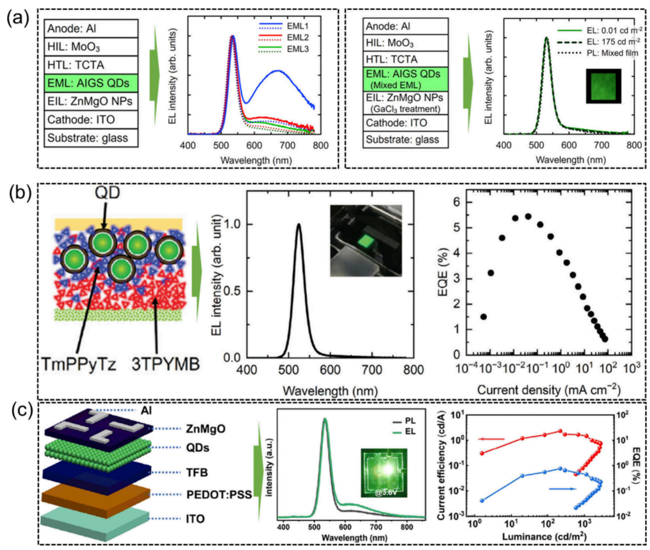

Figure 12. Electroluminescent performance of AIGS QLEDs employing (a) hybrid AIGS QDs as EML. Reproduced with permission from ref 90 Copyright 2023 American Chemical Society. (b) TmPPyTz-3TPYMB-AIGS QDs as EML. Reproduced with permission from ref 135 Copyright 2024 John Wiley & Sons, Ltd. (c) Pure AIGS QDs as EML. Reproduced with permission from ref 94. Copyright 2024 The Royal Society of Chemistry.

adding a second ETM of 3TPYMB in the QDs further boosted the EQE of the QLED to 5.4% (Figure 12b). This improvement was attributed to phase separation between the two ETMs, where each component occupied distinct regions and contributed differently to electron transport and recombination dynamics in the QDs EML. The synergistic effect of the ETM combination also led to narrower EL spectra with a fwhm of 30 nm (Figure 12b). Zheng et al. employed HF-treated AIGS/AGS QDs as the EML for QLEDs and studied the influence of HF treatment on their EL performance.[94] The EL spectrum in close resemblance to the PL of QDs was recorded, i.e., an EL peak at 535 nm with a moderately broad fwhm of 36 nm compared to 33 nm for the PL peak (Figure 12c), indicating effective suppression of both energy transfer between QDs and the quantum confinement Stark effect (QCSE) under an electric field. Notably, the EL intensity of the defect emission in the 600–800 nm range was enhanced relative to band-edge PL. The EL peak position remained stable with an increase in voltage, suggesting good emission stability under the electric field. A turn-on voltage of 2.2 V was achieved with a maximum luminance of 2747 cd m^{-2}. The highest current density observed was 1296 mA cm^{-2}, indicating efficient electron transfer from the ZnO ETL to QDs and enhanced charge recombination. The devices achieved a maximum EQE of 0.75%. The 0EL performance of the AIGS QLEDs is further summarized in Table 4.

Despite significant progress in AIGS QDs, their EL device performance remains constrained by surface-related defects,

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Table 4. Summary of the EL Performance of AIGS-Based QLEDs

No.	device structure	EL wavelength (nm)	peak luminance (cd/m^2)	EQE (%)	turn-on voltage (V)	ref
1	ITO/PEDOT:PSS/PTAA/QDs/TPBi/Al/LiF	535	47.8	0.60	2.8	28
2	ITO/PEDOT:PSS/TFB/(AIGS/AGS)/ZnMgO/Al	535	2747	0.75	2.2	94
3	ITO/ZnO/(AIGS/GaSx) + $GaCl_3$ 3TPYMB/TCTA/MoO_3/Al	531	175	1.1	2.4	90
4	ITO/ZnMgO/(AIGS/GaSx) + 3TPYMB or TmPPyT/TCTA/MoO_3/Al	524	1000	5.4	2	135

particularly Ga-deficient sites that act as trap states and reduce the radiative charge recombination efficiency. In addition, the nature of the shell composition plays a critical role, for example, an amorphous GaS_x shell often forms poor-quality interfaces with adjacent hole- and electron-transport layers, which severely hinders charge injection and transport in QLEDs. Together, these considerations highlight that future improvements will require coordinated optimization of the QD surface chemistry, shell design, and device-layer architecture to achieve higher efficiency, luminance, and color-pure emission for next-generation displays.

In conclusion, AIGS QDs have rapidly evolved from broad-emitting, defect-dominated materials into compositionally engineered, spectrally pure, and near-unity PLQY emitters, capable of directly competing with leading InP and Cd-based QDs. Recent breakthroughs, including halide-mediated synthesis, halide-induced interface reconstruction, defect-elimination strategies, and coherent AGS shell growth, have collectively reshaped the photophysical landscape of AIGS QDs. These advances have enabled narrow emission line widths (<35 nm), PLQYs exceeding 90%, and suppressed defect emission. These QDs have been successfully applied in the color conversion devices, where the performance achieved is superior to that of the InP-based counterpart. However, the EL devices fabricated from AIGS/GaS_x have shown extremely lower performance owing to the poor charge injection or instability of GaS_x-shelled AIGS QDs under high electric fields, suggesting that significant efforts are still required to unlock their potential in EL devices. The progress summarized in this review highlights not only the maturation of green AIGS QDs as a viable environment-benign emissive material but also their growing relevance in next-generation display and lighting technologies. The future developments seek efforts in the following areas:

Rational Heterostructure Design with Coherent Interface Engineering

Despite the remarkable achievements, several challenges remain before AIGS QDs can reach full commercial viability. A deeper mechanistic understanding of precursor conversion, Ga incorporation kinetics, defect formation pathways, and heterostructure stabilization is essential for a fully predictable and scalable synthesis. The development of unified synthetic principles for precise cation alloying, uniform shell growth, and reliable defect passivation remains a critical need to attain homogeneous AIGS cores. For AIGS QDs, although the GaS_x shell is very beneficial for attaining high PL outcomes, the disordered nature of such a shell makes it prone to degradation in the long term or during device fabrication. Therefore, the growth of the crystalline AGS shell should be preferred, and the growth parameters need to be optimized to ensure a coherent interface. For further enhancement of the photochemical stability, the growth of a wide band gap shell, such as ZnS or hybrid ZnGaS, on AIGS/AGS can be beneficial.

Ligand Passivation Strategies

A systematic exploration of ligand chemistry, involving X-, L-, and Z-type ligands, offers one of the most promising directions for advancing AIGS QDs beyond their current performance limits. Future studies should be focused on establishing clear structure−property relationships between ligand binding, surface stoichiometry, and defect suppression to achieve more predictable control over radiative and nonradiative recombination pathways. Beyond traditional long-chain ligands, the development of multidentate, dynamically binding ligands, and inorganic halide complexes may enable unprecedented passivation of both cationic and anionic surface vacancies. Such ligands could simultaneously tune the surface dipole, reduce intragap trap states, and stabilize Ga- or In-rich interfaces of bare QDs, leading to an efficient shell formation. Furthermore, smart postsynthesis ligand exchange protocols, designed to maintain colloidal stability while enhancing charge transport, are expected to play a crucial role in bridging the gap between high PLQY and high EQE in the EL devices.

Color Tunability of AIGS QDs

Although AIGS QDs are capable of emitting in a broad visible spectrum, the current research centers on the development of green-emitting composition only, and the blue- or red-emitting counterparts are less explored. This underscores the current limitations in the synthetic technologies of AIGS QDs. By leveraging the precursor chemistry of various metal−organic or metal halides in combination with coordinated or non-coordinated solvents, the synthesis conditions for blue and red emissive counterparts need to be developed. For the blue counterpart, as the band gap gets close to that of AGS, its implementation as a shell material may not be efficient in improving the optical properties owing to limited confinement of the excitons within the core; therefore, a wider band gap shell than AGS, retaining the band edge emission characteristics in blue AIGS QDs requires exploration. For red emissive AIGS, AGS still plays a crucial role in forming the type I band structure and will play a vital role in enhancing the optical properties.

Engineering the Device Structure

On the device side, rational engineering of charge transport and recombination pathways will be essential for pushing AIGS-based QLEDs toward high performance. Experiencing the current limitations in the EL performance of AIGS QLEDs, which mainly employ AIGS/GaS_x QDs mixed with ETMs as the EML for improved charge injection, remains a critical milestone for minimizing nonradiative losses and suppressing defect-assisted recombination. This discrepancy arises due to the detrimental role of the GaS_x-shell for charge injections owing to its amorphous nature, causing low-quality QDs-HTL/ETL interfaces. The halide-derived QDs are considered the better option for efficient QLEDs due to better charge transfer owing to shorter halide ligands. Thus, the halide-derived AIGS QDs with AGS shell should be employed as EML for QLED fabrication and the performance should be optimized. In addition, achieving improved charge injection through optimized electron/hole mobilities, as well as tailored transport layer energetics, is crucial for overcoming the charge injection limitations. Future progress is likely to come from interface-specific design strategies, including the use of self-assembled monolayers and dipole-engineered buffer layers that modulate band alignment while preserving the structural integrity of the AIGS-based QDs. Such interfaces can mitigate field-activated trap states, reduce Auger-induced quenching, and stabilize exciton formation within the EML.

Moreover, next-generation architectures integrating graded energy landscapes, multifunctional ETLs/HTLs, and hybrid inorganic–organic charge transport layers may enable more balanced carrier distribution and significantly enhance operational stability. Coupled with advanced patterning techniques and optical cavity engineering, these developments could unlock AIGS QLEDs with higher luminances and EQE.

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REFERENCES

(1) Han, T.-H.; Jang, K. Y.; Dong, Y.; Friend, R. H.; Sargent, E. H.; Lee, T.-W. A roadmap for the commercialization of perovskite light emitters. *Nature Reviews Materials* **2022**, *7* (10), 757−777.

(2) Channa, A. I.; Bai, S.; Wang, Z. M.; Tong, X. Advancements in Eco-Friendly Colloidal Quantum Dots and their Application in Light Emitting Diodes: Achieving Bright and Color-Pure Emission for Displays. *Laser & Photonics Reviews* **2025**, *19* (4), 2400678.

(3) Song, J.; Wang, O.; Shen, H.; Lin, Q.; Li, Z.; Wang, L.; Zhang, X.; Li, L. S. Over 30% External Quantum Efficiency Light-Emitting Diodes by Engineering Quantum Dot-Assisted Energy Level Match for Hole Transport Layer. *Adv. Funct. Mater.* **2019**, *29* (33), 1808377.

(4) Guan, J.; Ma, J.; Zhou, L.; Zou, Y.; Ye, L.; Hou, W.; Zhang, P.-k.; Dai, X.; Zhang, H.; Talapin, D. V.; Wang, Y. Direct ambient photopatterning of RGB quantum dots for light-emitting diodes with EQE exceeding 20%. *Nat. Commun.* **2025**, *16* (1), 9689.

(5) Qu, X.; Xiang, G.; Ma, J.; Liu, P.; Kyaw, A. K. K.; Wang, K.; Sun, X. W. Identifying the dominant carrier of CdSe-based blue quantum dot light-emitting diode. *Appl. Phys. Lett.* **2023**, *122* (11), 113501.

(6) Jang, E.; Kim, Y.; Won, Y.-H.; Jang, H.; Choi, S.-M. Environmentally Friendly InP-Based Quantum Dots for Efficient Wide Color Gamut Displays. *ACS Energy Letters* **2020**, *5* (4), 1316−1327.

(7) Click, S. M.; Rosenthal, S. J. Synthesis, Surface Chemistry, and Fluorescent Properties of InP Quantum Dots. *Chem. Mater.* **2023**, *35* (3), 822−836.

(8) Cui, Z.; Yang, D.; Qin, S.; Wen, Z.; He, H.; Mei, S.; Zhang, W.; Xing, G.; Liang, C.; Guo, R. Advances, Challenges, and Perspectives for Heavy-Metal-Free Blue-Emitting Indium Phosphide Quantum Dot Light-Emitting Diodes. *Advanced Optical Materials* **2023**, *11* (4), 2202036.

(9) Toufanian, R.; Chern, M.; Kong, V. H.; Dennis, A. M. Engineering Brightness-Matched Indium Phosphide Quantum Dots. *Chem. Mater.* **2021**, *33* (6), 1964−1975.

(10) Tamang, S.; Lincheneau, C.; Hermans, Y.; Jeong, S.; Reiss, P. Chemistry of InP Nanocrystal Syntheses. *Chem. Mater.* **2016**, *28* (8), 2491−2506.

(11) Tong, X.; Channa, A. I.; You, Y.; Wei, P.; Li, X.; Lin, F.; Wu, J.; Vomiero, A.; Wang, Z. M. Boosting the performance of eco-friendly quantum dots-based photoelectrochemical cells via effective surface passivation. *Nano Energy* **2020**, *76*, 105062.

(12) Wang, G.; Wei, H.; Shi, J.; Xu, Y.; Wu, H.; Luo, Y.; Li, D.; Meng, Q. Significantly enhanced energy conversion efficiency of CuInS2 quantum dot sensitized solar cells by controlling surface defects. *Nano Energy* **2017**, *35*, 17−25.

(13) Jalalah, M.; Al-Assiri, M. S.; Park, J.-G. One-Pot Gram-Scale, Eco-Friendly, and Cost-Effective Synthesis of CuGaS2/ZnS Nanocrystals as Efficient UV-Harvesting Down-Converter for Photovoltaics. *Adv. Energy Mater.* **2018**, *8* (20), 1703418.

(14) You, Y.; Tong, X.; Imran Channa, A.; Zhi, H.; Cai, M.; Zhao, H.; Xia, L.; Liu, G.; Zhao, H.; Wang, Z. High-efficiency luminescent solar concentrators based on Composition-tunable Eco-friendly Core/shell quantum dots. *Chemical Engineering Journal* **2023**, *452*, 139490.

(15) Wei, J.; Hu, Z.; Zhou, W.; Lu, H.; Zhang, W.; Guo, R. Color-converted white light-emitting diodes based on I-III-VI quantum dots: Package strategies and stability promotion. *Applied Materials Today* **2022**, *29*, 101585.

(16) Uematsu, T.; Wajima, K.; Sharma, D. K.; Hirata, S.; Yamamoto, T.; Kameyama, T.; Vacha, M.; Torimoto, T.; Kuwabata, S. Narrow band-edge photoluminescence from AgInS2 semiconductor nanoparticles by the formation of amorphous III−VI semiconductor shells. *NPG Asia Materials* **2018**, *10* (8), 713−726.

(17) Kameyama, T.; Kishi, M.; Miyamae, C.; Sharma, D. K.; Hirata, S.; Yamamoto, T.; Uematsu, T.; Vacha, M.; Kuwabata, S.; Torimoto, T. Wavelength-Tunable Band-Edge Photoluminescence of Nonstoichiometric Ag−In−S Nanoparticles via Ga3+ Doping. *ACS Appl. Mater. Interfaces* **2018**, *10* (49), 42844−42855.

(18) Yang, L.; Zhang, S.; Xu, B.; Jiang, J.; Cai, B.; Lv, X.; Zou, Y.; Fan, Z.; Yang, H.; Zeng, H. I−III−VI Quantum Dots and Derivatives: Design, Synthesis, and Properties for Light-Emitting Diodes. *Nano Lett.* **2023**, *23* (7), 2443−2453.

(19) Xie, X.; Jiang, T.; Lin, O.; Liu, J.; Zhang, Y.; Tang, A. Narrow-Bandwidth I−III−VI Semiconductor Nanocrystals: Synthesis, Luminescence and Applications in Quantum-Dot Light-Emitting Diodes. *Advanced Physics Research* **2024**, *3* (12), 2400071.

(20) Jin, L.; Selopal, G. S.; Tong, X.; Perepichka, D. F.; Wang, Z. M.; Rosei, F. Heavy-Metal-Free Colloidal Quantum Dots: Progress and Opportunities in Solar Technologies. *Adv. Mater.* **2024**, *36* (33), 2402912.

(21) Li, Z.; Channa, A. I.; Wang, Z. M.; Tong, X. Tailoring Eco-Friendly Colloidal Quantum Dots for Photoelectrochemical Hydrogen Generation. *Small* **2023**, *19* (50), 2305146.

(22) Rao, M. J.; Shibata, T.; Chattopadhyay, S.; Nag, A. Origin of Photoluminescence and XAFS Study of (ZnS)1−x(AgInS2)x Nanocrystals. *J. Phys. Chem. Lett.* **2014**, *5* (1), 167−173.

(23) Zaiats, G.; Kinge, S.; Kamat, P. V. Origin of Dual Photoluminescence States in ZnS−CuInS2 Alloy Nanostructures. *J. Phys. Chem. C* **2016**, *120* (19), 10641−10646.

(24) Jara, D. H.; Stamplecoskie, K. G.; Kamat, P. V. Two Distinct Transitions in CuxInS2 Quantum Dots. Bandgap versus Sub-Bandgap Excitations in Copper-Deficient Structures. *J. Phys. Chem. Lett.* **2016**, *7* (8), 1452−1459.

(25) Dana, J.; Debnath, T.; Ghosh, H. N. Involvement of Sub-Bandgap States in Subpicosecond Exciton and Biexciton Dynamics of Ternary AgInS2 Nanocrystals. *J. Phys. Chem. Lett.* **2016**, *7* (16), 3206−3214.

(26) Sarma, M.; Mondal, B.; Teotia, Y.; Adarsh, K. V.; Nag, A. Defect-Mediated Exciton Storage in Ag−In−Ga−S Nanocrystals. *ACS Energy Letters* **2025**, *10* (8), 3892−3899.

(27) Liu, G.; Xu, L.; Hu, Y.; Li, X.; Jeon, S.; Zhang, S.; Zeng, H. Unveiling the dual role of silver-associated defects: the manipulators of luminescence and carrier dynamics in eco-friendly AgIn0.5Ga0.5S2. *Science China Materials* **2024**, *67* (1), 214−222.

(28) Yan, D.; Chen, H.; Zhang, L.; Luo, Y.; Dong, Y.; Zou, Y.; Zeng, H. Intrinsic Defect Self-Compensation Enables Spectrum Narrowing of Uniform Alloyed Ag−In−Ga−S Quantum Dots. *ACS Energy Letters* **2025**, *10* (8), 3755−3762.

(29) Park, S.; Kim, B. J.; Jung, W. H.; Lee, K. H.; Lee, H.; Lim, J. Suppressing Tail Emission from AgIn1−xGaxS2/AgGaS2 Quantum Dots by GaI3−Assisted Interface Reinforcement. *ACS Nano* **2025**, *19* (29), 26831−26842.

(30) Mao, B.; Chuang, C.-H.; Wang, J.; Burda, C. Synthesis and Photophysical Properties of Ternary I−III−VI AgInS2 Nanocrystals: Intrinsic versus Surface States. *J. Phys. Chem. C* **2011**, *115* (18), 8945−8954.

(31) Lee, H. J.; Im, S.; Jung, D.; Kim, K.; Chae, J. A.; Lim, J.; Park, J. W.; Shin, D.; Char, K.; Jeong, B. G.; et al. Coherent heteroepitaxial growth of I-III-VI2 Ag(In,Ga)S2 colloidal nanocrystals with near-unity quantum yield for use in luminescent solar concentrators. *Nat. Commun.* **2023**, *14* (1), 3779.

(32) Uematsu, T.; Tepakidareekul, M.; Hirano, T.; Torimoto, T.; Kuwabata, S. Facile High-Yield Synthesis of Ag−In−Ga−S Quaternary Quantum Dots and Coating with Gallium Sulfide Shells for Narrow Band-Edge Emission. *Chem. Mater.* **2023**, *35* (3), 1094−1106.

(33) Sadovnikov, S. I.; Gusev, A. I. Recent progress in nanostructured silver sulfide: from synthesis and nonstoichiometry to properties. *Journal of Materials Chemistry A* **2017**, *5* (34), 17676−17704.

(34) Zeng, B.; Chen, F.; Liu, Z.; Guan, Z.; Li, X.; Teng, F.; Tang, A. Seeded-mediated growth of ternary Ag−In−S and quaternary Ag−In−Zn−S nanocrystals from binary Ag2S seeds and the composition-tunable optical properties. *Journal of Materials Chemistry C* **2019**, *7* (5), 1307−1315.

(35) Yang, J.; Ying, J. Y. Room-temperature synthesis of nanocrystalline Ag2S and its nanocomposites with gold. *Chem. Commun.* **2009**, No. 22, 3187−3189.

(36) Doh, H.; Hwang, S.; Kim, S. Size-Tunable Synthesis of Nearly Monodisperse Ag2S Nanoparticles and Size-Dependent Fate of the Crystal Structures upon Cation Exchange to AgInS2 Nanoparticles. *Chem. Mater.* **2016**, *28* (22), 8123−8127.

(37) Zhong, H.; Zhou, Y.; Ye, M.; He, Y.; Ye, J.; He, C.; Yang, C.; Li, Y. Controlled Synthesis and Optical Properties of Colloidal Ternary Chalcogenide CuInS2 Nanocrystals. *Chem. Mater.* **2008**, *20* (20), 6434−6443.

(38) Wang, Y.-H. A.; Zhang, X.; Bao, N.; Lin, B.; Gupta, A. Synthesis of Shape-Controlled Monodisperse Wurtzite CuInxGa1−xS2 Semiconductor Nanocrystals with Tunable Band Gap. *J. Am. Chem. Soc.* **2011**, *133* (29), 11072−11075.

(39) Choi, S.-H.; Kim, E.-G.; Hyeon, T. One-Pot Synthesis of Copper−Indium Sulfide Nanocrystal Heterostructures with Acorn, Bottle, and Larva Shapes. *J. Am. Chem. Soc.* **2006**, *128* (8), 2520−2521.

(40) Niezgoda, J. S.; Harrison, M. A.; McBride, J. R.; Rosenthal, S. J. Novel Synthesis of Chalcopyrite CuxInyS2 Quantum Dots with Tunable Localized Surface Plasmon Resonances. *Chem. Mater.* **2012**, *24* (16), 3294−3298.

(41) Perera, S. D.; Zhang, H.; Ding, X.; Nelson, A.; Robinson, R. D. Nanocluster seed-mediated synthesis of CuInS2 quantum dots, nanodisks, nanorods, and doped Zn-CuInGaS2 quantum dots. *Journal of Materials Chemistry C* **2015**, *3* (5), 1044−1055.

(42) Zhu, G.; Xu, Z. Controllable Growth of Semiconductor Heterostructures Mediated by Bifunctional Ag2S Nanocrystals as Catalyst or Source-Host. *J. Am. Chem. Soc.* **2011**, *133* (1), 148−157.

(43) Thi Thu Huong, T.; Loan, N. T.; Ung, T. D. T.; Tung, N. T.; Han, H.; Liem, N. Q. Systematic synthesis of different-sized AgInS2/GaSx nanocrystals for emitting the strong and narrow excitonic luminescence. *Nanotechnology* **2022**, *33* (35), 355704.

(44) Thanh, N. T. K.; Maclean, N.; Mahiddine, S. Mechanisms of Nucleation and Growth of Nanoparticles in Solution. *Chem. Rev.* **2014**, *114* (15), 7610−7630.
(45) Pejova, B.; do Nascimento, J. A.; Talbi, F.; Fawcett-Houghton, T. J.; Kerrigan, A.; Lari, L.; Douthwaite, R. E.; Pejov, L.; Lazarov, V. K. 2 eV band gap tuning and optical properties of AgIn5S8 quantum dots. *Nanoscale* **2025**, *17* (30), 17846−17861.
(46) Hoisang, W.; Uematsu, T.; Torimoto, T.; Kuwabata, S. Luminescent Quaternary Ag(InxGa1−x)S2/GaSy Core/Shell Quantum Dots Prepared Using Dithiocarbamate Compounds and Photoluminescence Recovery via Post Treatment. *Inorg. Chem.* **2021**, *60* (17), 13101−13109.
(47) Hoisang, W.; Uematsu, T.; Torimoto, T.; Kuwabata, S. Photoluminescence Stability Enhancement of Ag−In−Ga−S/GaSx Core/Shell Quantum Dots with Thicker Shells by the Addition of Gallium Diethyldithiocarbamate. *Chem. Lett.* **2021**, *50* (11), 1863−1866.
(48) Kwon, J.; Shin, Y.; Sung, Y.; Doh, H.; Kim, S. Silver Sulfide Nanocrystals and Their Photodetector Applications. *Accounts of Materials Research* **2024**, *5* (9), 1097−1108.
(49) Lin, L.-H.; Wu, C.-C.; Lai, C.-H.; Lee, T.-C. Controlled Deposition of Silver Indium Sulfide Ternary Semiconductor Thin Films by Chemical Bath Deposition. *Chem. Mater.* **2008**, *20* (13), 4475−4483.
(50) Jang, J. S.; Borse, P. H.; Lee, J. S.; Choi, S. H.; Kim, H. G. Indium induced band gap tailoring in AgGa1−xInxS2 chalcopyrite structure for visible light photocatalysis. *J. Chem. Phys.* **2008**, *128* (15), 154717.
(51) Yamato, K.; Iwase, A.; Kudo, A. Photocatalysis using a Wide Range of the Visible Light Spectrum: Hydrogen Evolution from Doped AgGaS2. *ChemSusChem* **2015**, *8* (17), 2902−2906.
(52) Huang, D.; Persson, C. Photocatalyst AgInS2 for active overall water-splitting: A first-principles study. *Chem. Phys. Lett.* **2014**, *591*, 189−192.
(53) Hollingsworth, N.; Roffey, A.; Islam, H.-U.; Mercy, M.; Roldan, A.; Bras, W.; Wolthers, M.; Catlow, C. R. A.; Sankar, G.; Hogarth, G.; et al. Active Nature of Primary Amines during Thermal Decomposition of Nickel Dithiocarbamates to Nickel Sulfide Nanoparticles. *Chem. Mater.* **2014**, *26* (21), 6281−6292.
(54) Manju, C. K.; Mohanty, J. S.; Sarkar, D.; Chennu, S.; Pradeep, T. Towards atomically precise luminescent Ag2S clusters separable by thin layer chromatography. *Journal of Materials Chemistry C* **2018**, *6* (21), 5754−5759.
(55) Yang, P.; Ando, M.; Taguchi, T.; Murase, N. Highly Luminescent CdSe/CdxZn1−xS Quantum Dots with Narrow Spectrum and Widely Tunable Wavelength. *J. Phys. Chem. C* **2011**, *115* (30), 14455−14460.
(56) Kelestemur, Y.; Shynkarenko, Y.; Anni, M.; Yakunin, S.; De Giorgi, M. L.; Kovalenko, M. V. Colloidal CdSe Quantum Wells with Graded Shell Composition for Low-Threshold Amplified Spontaneous Emission and Highly Efficient Electroluminescence. *ACS Nano* **2019**, *13* (12), 13899−13909.
(57) Uematsu, T.; Doi, T.; Torimoto, T.; Kuwabata, S. Preparation of Luminescent AgInS2−AgGaS2 Solid Solution Nanoparticles and Their Optical Properties. *J. Phys. Chem. Lett.* **2010**, *1* (22), 3283−3287.
(58) Park, J.; Joo, J.; Kwon, S. G.; Jang, Y.; Hyeon, T. Synthesis of Monodisperse Spherical Nanocrystals. *Angew. Chem., Int. Ed.* **2007**, *46* (25), 4630−4660.
(59) Kwon, S. G.; Hyeon, T. Formation Mechanisms of Uniform Nanocrystals via Hot-Injection and Heat-Up Methods. *Small* **2011**, *7* (19), 2685−2702.
(60) Talapin, D. V.; Lee, J.-S.; Kovalenko, M. V.; Shevchenko, E. V. Prospects of Colloidal Nanocrystals for Electronic and Optoelectronic Applications. *Chem. Rev.* **2010**, *110* (1), 389−458.
(61) Meyns, M.; Iacono, F.; Palencia, C.; Geweke, J.; Coderch, M. D.; Fittschen, U. E. A.; Gallego, J. M.; Otero, R.; Juárez, B. H.; Klinke, C. Shape Evolution of CdSe Nanoparticles Controlled by Halogen Compounds. *Chem. Mater.* **2014**, *26* (5), 1813−1821.
(62) Kwon, Y.; Oh, J.; Lee, E.; Lee, S. H.; Agnes, A.; Bang, G.; Kim, J.; Kim, D.; Kim, S. Evolution from unimolecular to colloidal-quantum-dot-like character in chlorine or zinc incorporated InP magic size clusters. *Nat. Commun.* **2020**, *11* (1), 3127.
(63) Zhang, W.; Tan, Y.; Duan, X.; Zhao, F.; Liu, H.; Chen, W.; Liu, P.; Liu, X.; Wang, K.; Zhang, Z.; Sun, X. W. High Quantum Yield Blue InP/ZnS/ZnS Quantum Dots Based on Bromine Passivation for Efficient Blue Light-Emitting Diodes. *Advanced Optical Materials* **2022**, *10* (15), 2200685.
(64) Choi, S.-W.; Kim, H.-M.; Yoon, S.-Y.; Jo, D.-Y.; Kim, S.-K.; Kim, Y.; Park, S. M.; Lee, Y.-J.; Yang, H. Aminophosphine-derived, high-quality red-emissive InP quantum dots by the use of an unconventional in halide. *Journal of Materials Chemistry C* **2022**, *10* (6), 2213−2222.
(65) Imran, M.; Paritmongkol, W.; Mills, H. A.; Hassan, Y.; Zhu, T.; Wang, Y.-K.; Liu, Y.; Wan, H.; Park, S. M.; Jung, E.; et al. Molecular-Additive-Assisted Tellurium Homogenization in ZnSeTe Quantum Dots. *Adv. Mater.* **2023**, *35* (45), 2303528.
(66) Thon, S. M.; Ip, A. H.; Voznyy, O.; Levina, L.; Kemp, K. W.; Carey, G. H.; Masala, S.; Sargent, E. H. Role of Bond Adaptability in the Passivation of Colloidal Quantum Dot Solids. *ACS Nano* **2013**, *7* (9), 7680−7688.
(67) Kim, Y.; Imran Channa, A.; Lee, Y.; Kong, Y.; Kim, H.-M.; Kim, Y.-H.; Min Park, S.; Kim, D.; Yang, H. Green Ag−In−Ga−S quantum dots as highly absorption-capable, efficient, and color-pure emitters. *Chemical Engineering Journal* **2024**, *486*, 150219.
(68) Pearson, R. G. Hard and soft acids and bases, HSAB, part 1: Fundamental principles. *J. Chem. Educ.* **1968**, *45* (9), 581.
(69) Suh, Y.-H.; Lee, S.; Jung, S.-M.; Bang, S. Y.; Yang, J.; Fan, X.-B.; Zhan, S.; Samarakoon, C.; Jo, J.-W.; Kim, Y.; et al. Engineering Core Size of InP Quantum Dot with Incipient ZnS for Blue Emission. *Advanced Optical Materials* **2022**, *10* (7), 2102372.
(70) Kim, Y.; Yeo, H.-J.; Park, S. M.; Ryu, S.; Kim, Y.-H.; Kim, H.-M.; Lee, C.; Kim, B.; Yang, H. Halide-mediated surface passivation and optical modulation in Ag−In−Ga−S quantum dots: The critical role of precursor selection. *Chemical Engineering Journal* **2025**, *514*, 163270.
(71) Lim, S. J.; Ma, L.; Schleife, A.; Smith, A. M. Quantum dot surface engineering: Toward inert fluorophores with compact size and bright, stable emission. *Coord. Chem. Rev.* **2016**, *320−321*, 216−237.
(72) Yin, Y.; Alivisatos, A. P. Colloidal nanocrystal synthesis and the organic−inorganic interface. *Nature* **2005**, *437* (7059), 664−670.
(73) Cossairt, B. M. Shining Light on Indium Phosphide Quantum Dots: Understanding the Interplay among Precursor Conversion, Nucleation, and Growth. *Chem. Mater.* **2016**, *28* (20), 7181−7189.
(74) Kowalik, P.; Bujak, P.; Penkala, M.; Maroń, A. M.; Ostrowski, A.; Kmita, A.; Gajewska, M.; Lisowski, W.; Sobczak, J. W.; Pron, A. Indium(II) Chloride as a Precursor in the Synthesis of Ternary (Ag−In−S) and Quaternary (Ag−In−Zn−S) Nanocrystals. *Chem. Mater.* **2022**, *34* (2), 809−825.
(75) Park, J.; Jayaraman, A.; Schrader, A. W.; Hwang, G. W.; Han, H.-S. Controllable modulation of precursor reactivity using chemical additives for systematic synthesis of high-quality quantum dots. *Nat. Commun.* **2020**, *11* (1), 5748.
(76) Li, Y.; Hou, X.; Shen, Y.; Dai, N.; Peng, X. Tuning the Reactivity of Indium Alkanoates by Tertiary Organophosphines for the Synthesis of Indium-Based Quantum Dots. *Chem. Mater.* **2021**, *33* (23), 9348−9356.
(77) Wei, N.; Zhu, H.; Yan, D.; Yang, S.; Xu, L.; Zhang, S.; Dong, Y.; Zou, Y.; Zeng, H. Reactivity-matched synthesis of monodisperse Ag(In,Ga)S2 QDs with efficient luminescence. *Nanoscale* **2024**, *16* (28), 13654−13662.
(78) Xie, R.; Rutherford, M.; Peng, X. Formation of High-Quality I−III−VI Semiconductor Nanocrystals by Tuning Relative Reactivity of Cationic Precursors. *J. Am. Chem. Soc.* **2009**, *131* (15), 5691−5697.
(79) Akkerman, Q. A.; Genovese, A.; George, C.; Prato, M.; Moreels, I.; Casu, A.; Marras, S.; Curcio, A.; Scarpellini, A.; Pellegrino, T.; et al. From Binary Cu2S to Ternary Cu−In−S and Quaternary

Cu−In−Zn−S Nanocrystals with Tunable Composition via Partial Cation Exchange. *ACS Nano* **2015**, *9* (1), 521−531.

(80) De Trizio, L.; Prato, M.; Genovese, A.; Casu, A.; Povia, M.; Simonutti, R.; Alcocer, M. J. P.; D'Andrea, C.; Tassone, F.; Manna, L. Strongly Fluorescent Quaternary Cu−In−Zn−S Nanocrystals Prepared from Cu1-xInS2 Nanocrystals by Partial Cation Exchange. *Chem. Mater.* **2012**, *24* (12), 2400−2406.

(81) Song, J.; Ma, C.; Zhang, W.; Li, X.; Zhang, W.; Wu, R.; Cheng, X.; Ali, A.; Yang, M.; Zhu, L.; et al. Bandgap and Structure Engineering via Cation Exchange: From Binary Ag2S to Ternary AgInS2, Quaternary AgZnInS alloy and AgZnInS/ZnS Core/Shell Fluorescent Nanocrystals for Bioimaging. *ACS Appl. Mater. Interfaces* **2016**, *8* (37), 24826−24836.

(82) Donegá, C. d. M. Synthesis and properties of colloidal heteronanocrystals. *Chem. Soc. Rev.* **2011**, *40* (3), 1512−1546.

(83) Zhao, H.; Rosei, F. Colloidal Quantum Dots for Solar Technologies. *Chem.* **2017**, *3* (2), 229−258.

(84) Sachanyuk, V. P.; Gorgut, G. P.; Atuchin, V. V.; Olekseyuk, I. D.; Parasyuk, O. V. The Ag2S−In2S3−Si(Ge)S2 systems and crystal structure of quaternary sulfides Ag2In2Si(Ge)S6. *J. Alloys Compd.* **2008**, *452* (2), 348−358.

(85) Tsuji, I.; Kato, H.; Kobayashi, H.; Kudo, A. Photocatalytic H2 Evolution Reaction from Aqueous Solutions over Band Structure-Controlled (AgIn)xZn2(1-x)S2 Solid Solution Photocatalysts with Visible-Light Response and Their Surface Nanostructures. *J. Am. Chem. Soc.* **2004**, *126* (41), 13406−13413.

(86) Smith, A. M.; Nie, S. Semiconductor Nanocrystals: Structure, Properties, and Band Gap Engineering. *Acc. Chem. Res.* **2010**, *43* (2), 190−200.

(87) Ivashchenko, I. A.; Danyliuk, I. V.; Olekseyuk, I. D.; Pankevych, V. Z.; Halyan, V. V. Phase equilibria in the quasiternary system Ag2S−Ga2S3−In2S3 and optical properties of (Ga55In45)2S300, (Ga54.59In44.66Er0.75)2S300 single crystals. *J. Solid State Chem.* **2015**, *227*, 255−264.

(88) Liu, Y.; Kim, D.; Morris, O. P.; Zhitomirsky, D.; Grossman, J. C. Origins of the Stokes Shift in PbS Quantum Dots: Impact of Polydispersity, Ligands, and Defects. *ACS Nano* **2018**, *12* (3), 2838−2845.

(89) Demchenko, D. O.; Wang, L.-W. Optical transitions and nature of Stokes shift in spherical CdS quantum dots. *Phys. Rev. B* **2006**, *73* (15), 155326.

(90) Motomura, G.; Uematsu, T.; Kuwabata, S.; Kameyama, T.; Torimoto, T.; Tsuzuki, T. Quantum-Dot Light-Emitting Diodes Exhibiting Narrow-Spectrum Green Electroluminescence by Using Ag−In−Ga−S/GaSx Quantum Dots. *ACS Appl. Mater. Interfaces* **2023**, *15* (6), 8336−8344.

(91) Yuan, C.; He, M.; Liao, X.; Liu, M.; Zhang, Q.; Wan, Q.; Qu, Z.; Kong, L.; Li, L. Interface defects repair of core/shell quantum dots through halide ion penetration. *Chemical Science* **2023**, *14* (45), 13119−13125.

(92) Kim, Y.; Ham, S.; Jang, H.; Min, J. H.; Chung, H.; Lee, J.; Kim, D.; Jang, E. Bright and Uniform Green Light Emitting InP/ZnSe/ZnS Quantum Dots for Wide Color Gamut Displays. *ACS Applied Nano Materials* **2019**, *2* (3), 1496−1504.

(93) Jo, J.-H.; Jo, D.-Y.; Lee, S.-H.; Yoon, S.-Y.; Lim, H.-B.; Lee, B.-J.; Do, Y. R.; Yang, H. InP-Based Quantum Dots Having an InP Core, Composition-Gradient ZnSeS Inner Shell, and ZnS Outer Shell with Sharp, Bright Emissivity, and Blue Absorptivity for Display Devices. *ACS Applied Nano Materials* **2020**, *3* (2), 1972−1980.

(94) Li, Z.; Cao, S.; Wang, K.; Li, Q.; Huang, Y.; Fu, H.; Zhao, J.; Yang, W.; Zheng, J. Highly efficient narrowed emitting AgInxGa1−xS2/AgGaS2 quantum dots via an HF-assisted one-pot synthesis strategy and their light-emitting diodes. *Journal of Materials Chemistry C* **2024**, *12* (18), 6528−6539.

(95) Kim, T.-G.; Zherebetskyy, D.; Bekenstein, Y.; Oh, M. H.; Wang, L.-W.; Jang, E.; Alivisatos, A. P. Trap Passivation in Indium-Based Quantum Dots through Surface Fluorination: Mechanism and Applications. *ACS Nano* **2018**, *12* (11), 11529−11540.

(96) Won, Y.-H.; Cho, O.; Kim, T.; Chung, D.-Y.; Kim, T.; Chung, H.; Jang, H.; Lee, J.; Kim, D.; Jang, E. Highly efficient and stable InP/ZnSe/ZnS quantum dot light-emitting diodes. *Nature* **2019**, *575* (7784), 634−638.

(97) Kawamoto, Y.; Ogura, K.; Shojiya, M.; Takahashi, M.; Kadono, K. F1s XPS of fluoride glasses and related fluoride crystals. *J. Fluorine Chem.* **1999**, *96* (2), 135−139.

(98) Palui, G.; Aldeek, F.; Wang, W.; Mattoussi, H. Strategies for interfacing inorganic nanocrystals with biological systems based on polymer-coating. *Chem. Soc. Rev.* **2015**, *44* (1), 193−227.

(99) Wu, L.; Hua, X.; Li, Y.; Zhang, Y.; Xue, X.; Deng, H.; Luo, Z.; Zhang, Y. Aggregation-based phase transition tailored heterophase junctions of AgInS2 for boosting photocatalytic H2 evolution. *J. Colloid Interface Sci.* **2022**, *628*, 721−730.

(100) Yang, Y.; Qin, H.; Peng, X. Intramolecular Entropy and Size-Dependent Solution Properties of Nanocrystal−Ligands Complexes. *Nano Lett.* **2016**, *16* (4), 2127−2132.

(101) Sluydts, M.; De Nolf, K.; Van Speybroeck, V.; Cottenier, S.; Hens, Z. Ligand Addition Energies and the Stoichiometry of Colloidal Nanocrystals. *ACS Nano* **2016**, *10* (1), 1462−1474.

(102) Pradhan, N.; Reifsnyder, D.; Xie, R.; Aldana, J.; Peng, X. Surface Ligand Dynamics in Growth of Nanocrystals. *J. Am. Chem. Soc.* **2007**, *129* (30), 9500−9509.

(103) Razgoniaeva, N.; Yang, M.; Garrett, P.; Kholmicheva, N.; Moroz, P.; Eckard, H.; Royo Romero, L.; Porotnikov, D.; Khon, D.; Zamkov, M. Just Add Ligands: Self-Sustained Size Focusing of Colloidal Semiconductor Nanocrystals. *Chem. Mater.* **2018**, *30* (4), 1391−1398.

(104) He, X.-H.; Yin, Y.-L.; Tang, G.; Liu, Y.; Jiang, F.-L. Role of Trioctylphosphine in the Synthesis of Quantum Dots: A Modulator of Nucleation, Growth, and Solubility. *J. Phys. Chem. C* **2023**, *127* (10), 5021−5028.

(105) Yoon, D.-E.; Yeo, S.; Lee, H.; Cho, H.; Wang, N.; Kim, G.-M.; Bae, W. K.; Lee, Y. K.; Park, Y.-S.; Lee, D. C. Pushing the Emission Envelope for Full-Color Realization of Colloidal Semiconductor Core/Shell Nanoplatelets. *Chem. Mater.* **2022**, *34* (20), 9190−9199.

(106) Nelson, A.; Ha, D.-H.; Robinson, R. D. Selective Etching of Copper Sulfide Nanoparticles and Heterostructures through Sulfur Abstraction: Phase Transformations and Optical Properties. *Chem. Mater.* **2016**, *28* (23), 8530−8541.

(107) Oh, N.; Shim, M. Metal Oleate Induced Etching and Growth of Semiconductor Nanocrystals, Nanorods, and Their Heterostructures. *J. Am. Chem. Soc.* **2016**, *138* (33), 10444−10451.

(108) Siy, J. T.; Bartl, M. H. Insights into Reversible Dissolution of Colloidal CdSe Nanocrystal Quantum Dots. *Chem. Mater.* **2010**, *22* (21), 5973−5982.

(109) Zhao, B.; Chen, L.; Liu, W.; Wu, L.; Lu, Z.; Cao, W. High efficiency blue light-emitting devices based on quantum dots with core-shell structure design and surface modification. *RSC Adv.* **2021**, *11* (23), 14047−14052.

(110) Wang, F.; Tang, R.; Kao, J. L. F.; Dingman, S. D.; Buhro, W. E. Spectroscopic Identification of Tri-n-octylphosphine Oxide (TOPO) Impurities and Elucidation of Their Roles in Cadmium Selenide Quantum-Wire Growth. *J. Am. Chem. Soc.* **2009**, *131* (13), 4983−4994.

(111) Dabbousi, B. O.; Rodriguez-Viejo, J.; Mikulec, F. V.; Heine, J. R.; Mattoussi, H.; Ober, R.; Jensen, K. F.; Bawendi, M. G. (CdSe)ZnS Core−Shell Quantum Dots: Synthesis and Characterization of a Size Series of Highly Luminescent Nanocrystallites. *J. Phys. Chem. B* **1997**, *101* (46), 9463−9475.

(112) Heuer-Jungemann, A.; Feliu, N.; Bakaimi, I.; Hamaly, M.; Alkilany, A.; Chakraborty, I.; Masood, A.; Casula, M. F.; Kostopoulou, A.; Oh, E.; et al. The Role of Ligands in the Chemical Synthesis and Applications of Inorganic Nanoparticles. *Chem. Rev.* **2019**, *119* (8), 4819−4880.

(113) Kovalenko, M. V.; Manna, L.; Cabot, A.; Hens, Z.; Talapin, D. V.; Kagan, C. R.; Klimov, V. I.; Rogach, A. L.; Reiss, P.; Milliron, D. J.; et al. Prospects of Nanoscience with Nanocrystals. *ACS Nano* **2015**, *9* (2), 1012−1057.

https://doi.org/10.1021/acsenergylett.6c00114
ACS Energy Lett. 2026, 11, 2647−2667

(114) Kirkwood, N.; Monchen, J. O. V.; Crisp, R. W.; Grimaldi, G.; Bergstein, H. A. C.; du Fossé, I.; van der Stam, W.; Infante, I.; Houtepen, A. J. Finding and Fixing Traps in II−VI and III−V Colloidal Quantum Dots: The Importance of Z-Type Ligand Passivation. *J. Am. Chem. Soc.* **2018**, *140* (46), 15712−15723.

(115) Paulson, P. D.; Dutta, V. Study of in situ CdCl2 treatment on CSS deposited CdTe films and CdS/CdTe solar cells. *Thin Solid Films* **2000**, *370* (1), 299−306.

(116) Panthani, M. G.; Kurley, J. M.; Crisp, R. W.; Dietz, T. C.; Ezzyat, T.; Luther, J. M.; Talapin, D. V. High Efficiency Solution Processed Sintered CdTe Nanocrystal Solar Cells: The Role of Interfaces. *Nano Lett.* **2014**, *14* (2), 670−675.

(117) Major, J. D.; Al Turkestani, M.; Bowen, L.; Brossard, M.; Li, C.; Lagoudakis, P.; Pennycook, S. J.; Phillips, L. J.; Treharne, R. E.; Durose, K. In-depth analysis of chloride treatments for thin-film CdTe solar cells. *Nat. Commun.* **2016**, *7* (1), 13231.

(118) Calvin, J. J.; Swabeck, J. K.; Sedlak, A. B.; Kim, Y.; Jang, E.; Alivisatos, A. P. Thermodynamic Investigation of Increased Luminescence in Indium Phosphide Quantum Dots by Treatment with Metal Halide Salts. *J. Am. Chem. Soc.* **2020**, *142* (44), 18897−18906.

(119) Yan, D.; Dong, Y.; Wei, N.; Yang, S.; Zhu, H.; Gu, W.; Zou, Y.; Zeng, H. High photoluminescence Ag-In-Ga-S quantum dots based on ZnX2-treated surface passivation. *Nano Research* **2024**, *17* (8), 7533−7541.

(120) Hoisang, W.; Uematsu, T.; Torimoto, T.; Kuwabata, S. Surface ligand chemistry on quaternary Ag(InxGa1−x)S2 semiconductor quantum dots for improving photoluminescence properties. *Nanoscale Advances* **2022**, *4* (3), 849−857.

(121) Long, R.; Chen, X.; Zhang, X.; Chen, F.; Wu, Z.; Shen, H.; Du, Z. Carboxylic-Free Synthesis of InP Quantum Dots for Highly Efficient and Bright Electroluminescent Device. *Advanced Optical Materials* **2023**, *11* (6), 2202594.

(122) Yoon, S.-Y.; Han, J.-N.; Lee, Y.-J.; Channa, A. I.; Jo, D.-Y.; Kim, H.-M.; Kim, Y.; Park, S. M.; Park, S.; Kim, Y.-H.; et al. Highly Emissive Green ZnSeTe Quantum Dots: Effects of Core Size on Their Optical Properties and Comparison with InP Counterparts. *ACS Energy Letters* **2023**, *8* (2), 1131−1140.

(123) Hahm, D.; Chang, J. H.; Jeong, B. G.; Park, P.; Kim, J.; Lee, S.; Choi, J.; Kim, W. D.; Rhee, S.; Lim, J.; et al. Design Principle for Bright, Robust, and Color-Pure InP/ZnSexS1−x/ZnS Heterostructures. *Chem. Mater.* **2019**, *31* (9), 3476−3484.

(124) Altıntas, Y.; Talpur, M. Y.; Ünlü, M.; Mutlugün, E. Highly Efficient Cd-Free Alloyed Core/Shell Quantum Dots with Optimized Precursor Concentrations. *J. Phys. Chem. C* **2016**, *120* (14), 7885−7892.

(125) Yu, P.; Cao, S.; Shan, Y.; Bi, Y.; Hu, Y.; Zeng, R.; Zou, B.; Wang, Y.; Zhao, J. Highly efficient green InP-based quantum dot light-emitting diodes regulated by inner alloyed shell component. *Light: Science & Applications* **2022**, *11* (1), 162.

(126) Chang, J. H.; Lee, H. J.; Rhee, S.; Hahm, D.; Jeong, B. G.; Nagamine, G.; Padilha, L. A.; Char, K.; Hwang, E.; Bae, W. K. Pushing the Band Gap Envelope of Quasi-Type II Heterostructured Nanocrystals to Blue: ZnSe/ZnSe1-XTeX/ZnSe Spherical Quantum Wells. *Energy Material Advances* **2021**, *2021*, na.

(127) Lee, S.-H.; Han, C.-Y.; Song, S.-W.; Jo, D.-Y.; Jo, J.-H.; Yoon, S.-Y.; Kim, H.-M.; Hong, S.; Hwang, J. Y.; Yang, H. ZnSeTe Quantum Dots as an Alternative to InP and Their High-Efficiency Electroluminescence. *Chem. Mater.* **2020**, *32* (13), 5768−5775.

(128) Huang, Z.; Sun, Q.; Zhao, S.; Wu, B.; Zhang, M.; Zang, Z.; Wang, Y. Deciphering Ultrafast Carrier Dynamics of Eco-Friendly ZnSeTe-Based Quantum Dots: Toward High-Quality Blue−Green Emitters. *J. Phys. Chem. Lett.* **2021**, *12* (49), 11931−11938.

(129) Xia, C.; Wu, W.; Yu, T.; Xie, X.; van Oversteeg, C.; Gerritsen, H. C.; de Mello Donega, C. Size-Dependent Band-Gap and Molar Absorption Coefficients of Colloidal CuInS2 Quantum Dots. *ACS Nano* **2018**, *12* (8), 8350−8361.

(130) Moreels, I.; Lambert, K.; De Muynck, D.; Vanhaecke, F.; Poelman, D.; Martins, J. C.; Allan, G.; Hens, Z. Composition and Size-Dependent Extinction Coefficient of Colloidal PbSe Quantum Dots. *Chem. Mater.* **2007**, *19* (25), 6101−6106.

(131) Karel Čapek, R.; Moreels, I.; Lambert, K.; De Muynck, D.; Zhao, Q.; Van Tomme, A.; Vanhaecke, F.; Hens, Z. Optical Properties of Zincblende Cadmium Selenide Quantum Dots. *J. Phys. Chem. C* **2010**, *114* (14), 6371−6376.

(132) Kamal, J. S.; Omari, A.; Van Hoecke, K.; Zhao, Q.; Vantomme, A.; Vanhaecke, F.; Capek, R. K.; Hens, Z. Size-Dependent Optical Properties of Zinc Blende Cadmium Telluride Quantum Dots. *J. Phys. Chem. C* **2012**, *116* (8), 5049−5054.

(133) Lu, H.; Hu, Z.; Zhou, W.; Wei, J.; Zhang, W.; Xie, F.; Guo, R. Synthesis and structure design of I−III−VI quantum dots for white light-emitting diodes. *Materials Chemistry Frontiers* **2022**, *6* (4), 418−429.

(134) Jo, D.-Y.; Yang, H. Synthesis of highly white-fluorescent Cu−Ga−S quantum dots for solid-state lighting devices. *Chem. Commun.* **2016**, *52* (4), 709−712.

(135) Motomura, G.; Ohisa, S.; Uematsu, T.; Kuwabata, S.; Kameyama, T.; Torimoto, T.; Fujisaki, Y. Pure Green Ag-In-Ga-S/Ga-S Quantum Dot Light-Emitting Diodes with Electron Transport Materials Exhibiting Enhanced Luminescence Properties. *Advanced Physics Research* **2024**, *3* (10), 2400042.

(136) Zhang, H.; Ma, X.; Lin, Q.; Zeng, Z.; Wang, H.; Li, L. S.; Shen, H.; Jia, Y.; Du, Z. High-Brightness Blue InP Quantum Dot-Based Electroluminescent Devices: The Role of Shell Thickness. *J. Phys. Chem. Lett.* **2020**, *11* (3), 960−967.

(137) Choi, M. K.; Yang, J.; Hyeon, T.; Kim, D.-H. Flexible quantum dot light-emitting diodes for next-generation displays. *npj Flexible Electronics* **2018**, *2* (1), 10.